

## Lay summary

A tiny jellyfish called *Turritopsis dohrnii* can do something no other animal can: when it gets old or sick or injured, it turns itself back into its own juvenile form and starts life over. Scientists think this trick must involve the same kinds of genes that control aging in every animal, including us. This project asked a simple question: which of the genes that matter for human aging and for human brain diseases does this jellyfish actually have?

The question turned out to be harder to answer than it should have been, for a surprising reason. When we looked up the jellyfish in the standard public gene databases, we found almost nothing: the databases have no proper gene records for this animal at all. That absence is itself a finding, because it means the world's most interesting longevity organism is nearly invisible to the tools most researchers would use first. So we bypassed the databases and searched the jellyfish's raw genome directly, using carefully checked methods, for two lists of genes: twenty-four genes at the heart of aging and longevity research, and twenty-four genes whose failure causes Alzheimer, Parkinson, ALS, and other brain diseases.

The result is a gene-by-gene map: where the assemblies show possible counterparts, copy-number signals, or no detectable strong hit in a fixed search. A matching sequence is a candidate to study, not evidence that the animal uses it to rejuvenate; a missing hit is not proof that the gene is absent. The possible extra *ATG5* copy could not be settled with the public long reads. Both halves matter. We also tested, carefully and with the statistics fixed in advance, whether the genes that connect aging and brain disease cluster together in this genome, and we report what we found whether or not it matched our hopes.

Everything was done with guardrails: the plan was written down before any data were looked at, the methods were borrowed unchanged from earlier validated work, positive controls had to reproduce known results, and fake scrambled genes were run through the same pipeline to prove it does not cry wolf. Where the evidence was weak, we say so. The aim was not to announce a miracle. It was to turn a famous biological marvel into a set of specific, checkable facts that aging researchers and brain researchers can actually use.

A later all-protein comparison did not validate a newly expanded family, and a three-stage RNA pilot was too sparse to show a reliable expression change. The new long-read test also did not settle ATG5 copy structure. These limits matter: this report is a transparent map and a set of testable leads, not a discovered cause of aging reversal or a proven new algorithm.

## **Introduction: the jellyfish that restarts its life, and what it owes to aging research and neurology**

Most animals are born, mature, reproduce, and age in one direction. *Turritopsis dohrnii*, a hydrozoan jellyfish barely five millimetres across, is the only known metazoan that can run this sequence backwards. When a medusa of this species is starved, injured, or simply old, it does not have to die. It sinks to the seabed, retracts its tentacles, loses the differentiated medusa form, and rebuilds a polyp, the juvenile stage of the life cycle, from its existing cells. That polyp can bud off new medusae that are genetic copies of the original. In principle the loop can repeat indefinitely, which is why the popular press named it the immortal jellyfish and why aging researchers have watched it with a mixture of envy and suspicion ever since the life cycle reversal was first described in detail.

The biological interest is not the immortality headline. It is the mechanism. Life cycle reversal in *T. dohrnii* is a whole-organism act of cellular rejuvenation. A differentiated adult cell population is induced to abandon its specialised identity and reacquire a younger developmental state, after which the organism rebuilds itself. In the vocabulary of modern biomedicine this is *in vivo* reprogramming without an experimenter, and it sits conceptually next to the two most discussed technologies in aging research: partial reprogramming by forced expression of pluripotency factors, and the still poorly understood ability of some regeneration-competent animals to rebuild damaged or missing tissue without scarring or loss of pattern. If a five-millimetre cnidarian can reboot its entire body repeatedly, then rejuvenation is not forbidden by the laws of metazoan biology; it is a trait that evolution has produced at least once, and traits that evolution has produced can, in principle, be studied, mechanistically decomposed, and eventually imitated.

That is the starting point of this project. The user directive behind this expansion was explicit: the jellyfish research must improve, and it must connect to other major problems, naming aging and longevity research and neurology. Those connections are not decorative. The first is direct:

rejuvenation is the inverse of aging, so the genome of a rejuvenating animal is a natural experiment in the genetics of age resistance, and the genes it shares with human longevity pathways are candidate levers for healthy lifespan extension. The second is less obvious but equally strong. The adult human brain is the organ least able to replace its cells; neurodegenerative disease is, in large part, the slow failure of maintenance biology in long-lived post-mitotic neurons. A creature whose nervous system dissolves and reforms during rejuvenation is a creature whose neurons tolerate a degree of plasticity that human neurons never experience. Understanding what makes that tolerance possible speaks directly to the two great unmet needs of neurology: stopping neurodegeneration, and inducing regeneration where degeneration has already occurred.

This document is the expansion report of MEGA27 item 26b. The closed phase of the project established two things. First, an audit of public databases showed that the standard gene-level resources (NCBI nucleotide, gene, and protein databases, and UniProt) contain essentially no curated gene records for *T. dohrnii*: for an eighteen-gene DNA-repair and telomerase panel, the count of gene-level records was zero, and only raw genome assemblies existed. That annotation desert is itself a result about the state of public biology. Second, the closed phase built a working translated-homology mapping pipeline (tBLASTn with frozen thresholds, locus clustering, and locus extraction) and demonstrated that the panel genes are in fact physically present and mappable in the two available *T. dohrnii* genome assemblies, with a curated locus ledger as the deliverable. The closed phase therefore ended with a calibrated boundary: the genes exist in the genome but are invisible to the databases that most computational workflows would consult first.

The expansion takes the next three steps. It widens the gene panel from DNA repair and telomere maintenance to two new, pre-registered panels: twenty-four aging and longevity genes chosen from the core of the hallmarks-of-aging and GenAge literature (Panel B), and twenty-four neurology and neurodegeneration genes comprising the major monogenic disease genes and neurotrophic factors (Panel C). It adds a new comparator genome, the chromosome-level assembly of *Turritopsis rubra*, the non-reverting congener that anchors the natural contrast between a rejuvenating and a non-rejuvenating species. And it applies the frozen, already-validated mapping pipeline, unchanged, with positive and negative controls, so that every

presence or absence claim in the new panels inherits the calibration of the closed phase rather than resting on fresh, untuned judgement.

Everything in this expansion is pre-registered. The analysis plan, the panel gene lists, the hypotheses H1 through H6, the statistical tests, the multiple-testing correction, and the honest-negative register were committed to the repository before any outcome data were fetched. If the aging panel fails to map, that is reported. If the neurology panel shows no enrichment, that is reported. If the copy-number contrast between the immortal and the mortal *Turritopsis* species evaporates under correction, that is reported. The purpose of the project is not to manufacture a discovery; it is to find out what is actually in the genome of the animal that restarts its life, and to route whatever is found, positive or negative, into the two fields that the user asked to see connected: the science of aging and longevity, and the science of the degenerating and regenerating brain.

A note on scope and honesty is warranted here, because the word immortality invites overclaiming. Nothing in this work shows that any human disease gene has a functional ortholog in the jellyfish, let alone that the jellyfish uses it the way humans do. Nothing here demonstrates that a jellyfish longevity gene would extend human healthspan. What this work does establish, gene by gene and genome by genome, is the substrate: which parts of the human aging and neurological-disease machinery have ancient, mappable counterparts in a rejuvenating cnidarian, which parts are absent, and where the copy-number and sequence signals differ between the immortal species and its mortal relatives. Mechanistic claims belong to the laboratory. This is a computational atlas, and it is written to be useful as one.

The structure of the report follows the logic of the work. Chapter 2 summarises the closed phase and the state of public resources for this organism. Chapter 3 develops the aging and longevity connection in depth, from the hallmarks framework to the comparative genomics of long-lived species. Chapter 4 develops the neurology connection, from the genetics of neurodegeneration to the peculiar biology of the cnidarian nervous system. Chapter 5 describes the expansion methods, including the controls and the statistical protocol. Chapters 6 and 7 report the mapping results for the aging and neurology panels gene by gene. Chapter 8 reports the pre-registered hypothesis tests, including the ones that failed. Chapter 9 discusses what the results mean for aging and longevity research, and Chapter 10 does the same for neurology. Chapter 11 collects

the honest negatives and the limitations, and Chapter 12 concludes with the most defensible next experiments.

The framing of this project follows an archetype that has succeeded at the top level of student science fairs: identify a biomarker where none exists, verify it through orthogonal lines of evidence, and state a concrete translational path. The template case is Natasha Kulviwat's Regeneron ISEF 2023 project, which won the Gordon E. Moore Award for nominating claudin-5 as a novel biomarker of suicide pathogenesis: human postmortem brain tissue stratified by cause of death supplied the primary evidence, ELISA, immunolocalization, and public RNA-sequencing differential-expression and pathway analyses supplied orthogonal verification, and molecular docking of existing drugs against the biomarker supplied the translational arm (Society for Science project 23691). This project applies the same structure at the genomic level. The exploratory candidates are named loci whose detection or apparent copy-number pattern can be tested across assemblies and relatives; none is a validated biomarker of rejuvenation. The orthogonal verification arms are assembly replication, negative controls, annotated-genome calibration, and panel-versus-panel contrasts. The translational perspective is a literature connection to human aging and neurodegeneration biology; it neither establishes conserved function nor verifies a therapeutic target in this species. What the archetype demands, and what the results chapters therefore deliver, is a named top candidate with a falsifiable validation plan rather than statistics alone.

## **Background: what the closed phase established, and where the boundary lay**

The organism and its trick

*Turritopsis dohrnii* is a hydrozoan in the family *Oceaniidae*, originally described from the Mediterranean and now known to be widely distributed in warm and temperate seas. Like most hydrozoans it has a biphasic life cycle: a benthic, colonial polyp stage buds off free-swimming medusae, which grow, reproduce sexually, and in every other known medusozoan species, die. The defining observation for *T. dohrnii*, built up through the 1990s and consolidated in the literature since, is that its medusae respond to stress, starvation, or senescence by reverting to the polyp stage. The medusa settles, its tissues reorganise, and a stolon and then a polyp emerge. The transformation involves transdifferentiation and, in some tissues, proliferation of a reactivated

stem-like cell population. The polyp then grows and buds new medusae, completing a loop that has no obligatory endpoint.

Two features of this biology matter for everything that follows. The first is that rejuvenation is whole-body and includes the nervous system. The medusa has a nerve ring, ocelli, and differentiated neural cell types; the polyp has a simpler but genuine nerve net. During reversal the medusa's neural structures are dismantled and a new nervous system is built. Whatever molecular machinery permits that degree of neural plasticity is operating in a living, intact, evolutionarily successful animal, not in a dish. The second feature is that reversal is not immortality in the literal sense: individual animals can still be eaten, infected, or fatally damaged, and the efficiency of reversal declines with repeated stress in laboratory settings. The accurate statement is that *T. dohrnii* lacks an obligatory senescent endpoint for the medusa stage, and that this is already strange enough to demand a genomic explanation.

The reference point: comparative genomics of mortal and immortal cnidarians

The field's anchor is the 2022 PNAS study by Pascual-Torner and colleagues, which sequenced and compared the genomes of *T. dohrnii* and its non-reverting congener *T. rubra*, and profiled the transcriptome of the life cycle reversal process itself. The headline findings were variants and expansions of genes associated with replication, DNA repair, telomere maintenance, redox environment, stem-cell population maintenance, and intercellular communication in the immortal species, together with silencing of polycomb repressive complex 2 targets and activation of pluripotency targets during reversal. The interpretation offered was that rejuvenation capacity rests on an unusually well-equipped genome maintenance and reprogramming toolkit.

That paper is both the motivation and the caution for the present project. The motivation is obvious: if repair, telomere, and stemness genes are expanded or diverged in the immortal species, a systematic gene-by-gene census of those systems is directly informative. The caution is subtler. Expansions and variants reported from a two-species comparison are hypotheses about mechanism, not proofs of it, and they depend entirely on the quality of the underlying annotation. If the annotation is incomplete or inaccessible, the comparison cannot be reproduced, and a claim that cannot be reproduced cannot be trusted or extended. That is exactly where the closed phase of this project found the public resources to be.

The annotation desert, measured

The closed phase of item 26b began with a simple plan: fetch the eighteen genes of a curated DNA-repair and telomerase panel for *T. dohrnii* and the moon jelly *Aurelia aurita* from the standard public databases, and compare them against human and *Hydra vulgaris* references. The plan failed in an instructive way. For *T. dohrnii*, systematic queries of the NCBI nucleotide, gene, and protein databases, with synonym expansion including the older name *Turritopsis nutricula*, returned zero gene-level records for every gene in the panel. UniProt held only a handful of *T. dohrnii* entries, none in the panel. The same pattern held for *A. aurita*. The only usable public material was raw genome assemblies: a contig-level assembly (GCA\_027922465.2) and the scaffold-level assembly associated with the Oviedo group (GCA\_025167195.1), plus the *A. aurita* assembly GCA\_004194415.1.

This finding, documented quantitatively in the closed phase report, was named the annotation desert: the organisms most interesting for longevity research are among the least served by the curated resources that computational biology defaults to. The practical consequence is that any researcher who begins, as most would, with a gene database query concludes incorrectly that the genes are absent. Absence of annotation is not absence of genes. The closed phase proved the distinction in the only rigorous way available: it bypassed the databases, took the raw genomes, and mapped the genes directly.

The calibrated pipeline of the closed phase

The mapping pipeline that the closed phase validated, and that this expansion reuses unchanged, is deliberately simple and auditable. Curated query proteins, human RefSeq references plus cnidarian homologs from annotated species (*Clytia hemisphaerica*, *Hydra vulgaris*, *Acropora*, *Nematostella*), are searched against each genome with translated BLAST (tBLASTn 2.17.0, e-value threshold  $1e-5$ ). Hits are clustered into candidate loci on single contigs within a fixed span, each locus accumulates a total bit score across its high-scoring segment pairs, and a locus is called strong when its total bit score reaches a frozen threshold of one hundred. Strong loci are extracted with flanks for downstream analysis. The thresholds were fixed before the closed phase's outcome analyses and are reused here verbatim; the expansion changes the queries and adds one comparator genome, nothing else.

On this pipeline the closed phase established that the repair and telomerase panel is physically present in the *T. dohrnii* genome. The locus ledger records, for example, a high-scoring FEN1 locus with nearly ninety percent query coverage and strong matches for MLH1 and other panel genes, in both *T. dohrnii* assemblies. A dedicated telomerase deep-check found the catalytic TERT locus, with its characteristic reverse-transcriptase and telomerase RNA-binding domains, in all three scanned genomes, including the mortal *A. aurita*, a result that immediately complicates any naive story in which telomerase presence alone explains immortality. A separate arm showed that the telomerase RNA component TERC, which is not a protein and therefore invisible to translated search, could not be detected by short-sequence alignment in the *T. dohrnii* assemblies, an honest negative that delimits what the method can and cannot see. Machine-learning arms (a convolutional classifier and a graph neural network over sequence composition) were trained and evaluated with an honest-negative register; the graph model did not beat baseline, and that was reported as such.

Why the expansion starts from genomes, not databases

The combined lesson of the closed phase is the methodological premise of this expansion: for organisms in the annotation desert, genome-first analysis is not a preference but a necessity. Every claim in the chapters that follow is therefore built on direct genome mapping with calibrated thresholds, positive controls that reproduce the closed phase's calls, and negative controls built from shuffled query sequences. The databases are used for what they are genuinely good at, which is supplying well-curated query proteins from annotated reference species, and for nothing more. Where the expansion extends beyond the closed phase, it does so by adding pre-registered panels and a new comparator genome, not by relaxing the evidentiary standard. The boundary the closed phase found, between what is annotated and what is present, is the boundary this expansion was built to cross.

## The annotation desert, revisited

The closed phase made a strong claim and this expansion depends on it being understood correctly, so it deserves a chapter of its own. The claim was never that *T. dohrnii* lacks genes. It was that the public gene-level infrastructure lacks *T. dohrnii*. The distinction sounds pedantic until you price it. A researcher who trusts the standard databases sees an empty organism: no

gene records, no curated proteins, nothing to download for the eighteen repair and telomerase genes that the longevity literature says to look at first. A researcher who goes to the raw assemblies finds the genes sitting there, unmappable by database search because nobody has deposited the annotation. The same organism is empty and full at once, depending entirely on which door you knock on.

This expansion priced the distinction precisely. On the database side, the audit stands as the closed phase measured it: zero gene-level records for the panel in NCBI nucleotide, gene, and protein for both *T. dohrnii* and *A. aurita*, near-zero UniProt coverage, synonym search exhausted. On the genome side, the closed phase's own ledger and this expansion's positive control now quantify the other half: the panel genes map, with strong loci, in the very genome assemblies the databases have nothing to say about. The annotation desert is therefore not a metaphor but a measured gap between two public faces of the same data: the sequence archives, which hold the raw genomes, and the curated gene resources, which have not caught up.

Why does the gap persist? Partly throughput: genome sequencing industrialised faster than genome annotation, and the long tail of non-model organisms is sequenced in draft form far faster than curators and pipelines can annotate. Partly economics: annotation effort follows biomedical funding, and a hydrozoan with no fishery, no pathogen role, and no model-organism community loses out to yet another mammal. And partly structure: automatic annotation propagates from existing annotations, so organisms distant from the annotated core get the worst predictions, which discourages deposition, which keeps them distant. The desert is self-reinforcing.

The consequence for aging and neurology research is the part this project cares most about. Negligibly senescent and rejuvenating species are disproportionately non-model: they are interesting precisely because they are not the standard laboratory animals, which means they are precisely the organisms the curated databases serve worst. Comparative longevity research therefore lives in the annotation desert by necessity, and its methods must be desert-proof: genome-first, calibration-first, threshold-frozen, controls-always. The pipeline the closed phase built and this expansion reuses is one such method, and it is deliberately mundane: BLAST, a fixed bit-score threshold, locus clustering, controls. Nothing in it is clever, which is why it can be trusted where cleverer methods would demand trust in their tuning.

There is also a quiet opportunity in the desert. Because the curated resources are empty, every gene this project maps is, in a small way, a deposition-quality finding: a specific locus, on a specific contig, with specific coordinates, identity, coverage, and evidence, recorded in a public ledger. Aggregated across the three panels, the expansion's ledgers amount to a partial first-pass annotation of the maintenance genome of *T. dohrnii*, produced as a side effect of asking the research questions. If the field ever builds the proper annotation, these ledgers are scaffolding for it. Until then, they are the difference between knowing the genes are there and merely assuming the 2022 study's genome must contain them.

## The aging and longevity connection

Why a rejuvenating jellyfish belongs in aging research

Aging research has a standing problem: it is very good at describing what goes wrong in old animals and much worse at explaining why some animals barely age at all. The hallmarks framework, most recently expanded by Lopez-Otin and colleagues in 2023, catalogues the recurring molecular damage and dysregulation of aging: genomic instability, telomere attrition, epigenetic alteration, loss of proteostasis, disabled macroautophagy, deregulated nutrient sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion, altered intercellular communication, chronic inflammation, and dysbiosis. The framework is descriptive and mammal-centric. It tells us what a failing maintenance system looks like. It does not tell us what an exceptionally successful one looks like, because the animals it is built on all fail eventually.

Negligibly senescent and rejuvenating organisms are the natural counterweight. Hydra, a freshwater cnidarian relative of *T. dohrnii*, shows no detectable increase in mortality with age under laboratory conditions, a result established by Martinez in the 1990s and refined by Schaible and colleagues in 2015 with constant mortality and fertility across years of observation. The bowhead whale lives more than two centuries and, as a 2025 Nature study showed, carries quantitatively better DNA double-strand-break repair than shorter-lived mammals. The naked mole rat combines negligible senescence with extraordinary cancer resistance, and comparative genomic work across long-lived mammals repeatedly converges on the same functional neighbourhoods: genome maintenance, telomere biology, proteostasis, nutrient sensing, and

tumour suppression. *T. dohrnii* is the extreme member of this club. It does not merely resist aging; it reverses its effects through a developmentally programmed rejuvenation. If comparative longevity research is serious about learning from the winners, the immortal jellyfish is not an exotic footnote but the limiting case.

The logic of the comparison is what makes Panel B of this expansion necessary. The closed phase mapped the repair and telomerase machinery. But rejuvenation is presumably not only a repair phenomenon. The transcriptomics of life cycle reversal and the Pascual-Torner comparative genomics both point at pluripotency control, redox handling, and intercellular signalling. A rigorous census therefore has to ask, for each of the canonical aging-pathway genes, whether the immortal jellyfish has it at all, whether its copy number or sequence diverges from its mortal relatives, and whether the pattern across genes resembles the pattern in other long-lived species. That is exactly what Panel B was pre-registered to measure.

Panel B: the twenty-four genes and why each is in the longevity canon

The aging and longevity panel was frozen before any data were fetched. Its membership is not arbitrary; each gene is a load-bearing element of the longevity literature, and the panel deliberately spans the major mechanistic categories so that the presence map reads as a profile of maintenance strategies rather than a cherry-picked list.

The insulin and IGF signalling module is the most conserved longevity pathway in the animal kingdom. Reduced signalling through the insulin receptor, IGF1 receptor, and their downstream kinase AKT extends lifespan in worms, flies, and mice, and the same axis implicates human longevity through FOXO3, the forkhead transcription factor whose protective alleles are among the most replicated findings in human longevity genetics, and through the growth hormone receptor GHR, whose loss produces the long-lived, dwarfism-associated Laron syndrome. PTEN, the lipid phosphatase that opposes this pathway, and IGF1 itself complete the module. The nutrient-sensing module continues with MTOR, the kinase at the heart of rapamycin-sensitive growth control and the single most pharmacologically validated aging target, and PRKAA1, the catalytic subunit of AMPK, the energy-depletion sensor whose activation is linked to metformin-like longevity effects.

The sirtuins SIRT1 and SIRT6 represent the NAD-dependent deacetylase family that couples metabolism to chromatin state and DNA repair; SIRT6 in particular has direct lifespan effects in mice and sits at the crossroads of repair and metabolism. The genome-guardian module contains TP53, the master regulator of the DNA-damage response whose calibrated activity balances cancer suppression against stem-cell attrition; CDKN2A, encoding the p16 senescence brake whose expression is one of the most widely used biomarkers of aging; the checkpoint kinases ATM and ATR; and the RecQ helicases BLM and WRN, whose loss produces the human progeroid Bloom and Werner syndromes. These last two are the strongest human genetic evidence that helicase-driven genome maintenance is a rate-limiting aging system: break the helicase, get premature aging.

The proteostasis and quality-control module contains NFE2L2, encoding the NRF2 transcription factor that orchestrates the oxidative-stress response and is recurrently implicated in longevity; HSF1, the heat-shock factor that maintains the folding environment and declines with age; the autophagy core genes ATG5 and ATG7, whose loss shortens lifespan and whose induction extends it in multiple models; BECN1, the beclin that initiates autophagosome formation; and TFEB, the master transcriptional regulator of lysosomal biogenesis, whose overexpression clears aggregates and extends healthspan in model systems. Finally, APOE sits deliberately on the boundary of Panels B and C: its alleles are the strongest common genetic influence on both human longevity and Alzheimer disease risk, making it the single clearest molecular bridge between the two fields the user asked to connect.

What the jellyfish can and cannot tell us about these genes

Three outcomes are possible for each Panel B gene, and all three are informative. A gene may be present and conserved in *T. dohrnii*, which identifies it as ancient metazoan maintenance machinery and makes its handling in a rejuvenating animal a legitimate mechanistic question: does the immortal jellyfish use the same gene differently? A gene may be present but expanded or diverged relative to *T. rubra* and *A. aurita*, which is the signature the Pascual-Torner study reported for repair genes and which, if it recurs in the longevity panel, nominates specific maintenance systems as candidate rejuvenation adaptations. Or a gene may be absent from the jellyfish genome within the calibrated limits of the method, which is equally meaningful: it would show that a recognisable version of that maintenance pathway is dispensable for

biological immortality, and therefore that the pathway's role in human aging is a derived feature of our lineage rather than a universal constraint.

The copy-number contrast deserves a special caveat that this report will keep visible. The immortal-versus-mortal comparison here runs across four genomes: two *T. dohrnii* assemblies, one *T. rubra*, one *A. aurita*. That is a powerful natural design for generating hypotheses and a weak one for proving adaptation, because every contrast is confounded with assembly quality, annotation state, and the deep phylogenetic distance between the genera. The pre-registered H3 and H4 tests are therefore written as screens, and their positive results are nominations, not discoveries. The report holds to that discipline even where the results are pleasing.

Connection to the wider longevity-genomics record

Placed against the mammalian longevity literature, the jellyfish census plays a distinctive role. Mammalian studies can compare long-lived and short-lived species that are all, ultimately, aging species; their contrasts are quantitative. The *Turritopsis* comparison offers a qualitative one: mortality endpoint versus no obligatory mortality endpoint, inside the same genus. If a maintenance gene is expanded in the immortal species relative to its mortal congener, that is a cleaner signal than most cross-order mammalian comparisons can provide, because *T. dohrnii* and *T. rubra* share ecology, body plan, and most of their genome. The cost is statistical power: one genus, one contrast. The benefit is specificity. This is the trade the expansion makes openly, and it is why every positive contrast is paired with an explicit statement of what would be needed to confirm it: orthogonal evidence from independent assemblies, transcriptomics across the reversal timecourse, and eventually functional perturbation, which is now feasible in cnidarians by CRISPR.

The deliverable for aging research, then, is a prioritised, falsifiable map: which longevity genes the immortal jellyfish keeps, which it amplifies, and which it does without. Each row of that map is a small, testable claim about the necessary and dispensable parts of the maintenance toolkit, generated by an animal that already solved the problem the field is trying to solve.

## The neurology connection

Two failures that define modern neurology

Neurology's two great unsolved problems are really one problem seen from two directions. Neurodegeneration is the slow loss of neurons that were built to last a lifetime and are not replaced: dopaminergic neurons in Parkinson disease, motor neurons in ALS, cortical and hippocampal neurons in Alzheimer disease and the frontotemporal dementias. Failed regeneration is the mirror image: after stroke, trauma, or degeneration, the adult human nervous system does not rebuild, because adult human neurons are post-mitotic, their progenitors are nearly absent, and the maintenance systems that keep them alive degrade with age. Both failures are failures of cellular maintenance over long timescales. Both are exactly the class of problem that a repeatedly rejuvenating nervous system does not have.

The cnidarian nervous system is ancient, real, and rebuildable

It is tempting to dismiss the jellyfish nervous system as too simple to matter for human neurology. That would be a mistake, for three reasons. First, it is genuinely neural: cnidarian neurons use the same neurotransmitter families, the same electrical excitability, and homologs of the same developmental transcription factors as bilaterian neurons, and the cnidarian-bilaterian common ancestor already possessed a differentiated nervous system, making neural cell types among the oldest cell types in animals. Comparative work on cnidarian neurogenesis has been explicit that these animals preserve, in readable form, ancestral features of how nervous systems are made. Second, it is continuously rebuilt. In *Hydra*, the nerve net is perpetually renewed from interstitial stem cells across the animal's non-aging life. In *T. dohrnii*, the entire neural architecture of the medusa is dismantled and re-specified during life cycle reversal. Third, it is experimentally tractable: cnidarian genomes are small, their cell lineages are accessible, and gene editing now works in several species.

The relevance to human neurology is not that a jellyfish has a brain. It is that the jellyfish genome encodes the ancestral version of the maintenance and plasticity machinery that human neurons depend on and cannot renew. Where that machinery exists in both species, the jellyfish offers a context in which the machinery runs in a permanently regenerative mode. Where the human disease genes have no jellyfish counterpart, the comparison marks the parts of human neural vulnerability that are evolutionary novelties, layered on top of the ancient core. Both readings are useful, and Panel C was pre-registered to produce the gene-by-gene map.

Panel C: the twenty-four genes and what each costs when it fails

The neurology panel is built from the genes whose failure causes or strongly predisposes to the major human neurodegenerative diseases, plus the neurotrophic signalling system. The Alzheimer module contains APP, the amyloid precursor protein whose cleavage products drive the defining pathology; PSEN1 and PSEN2, the presenilins that form the catalytic core of the gamma-secretase that cuts APP and whose mutations cause aggressive early-onset familial disease; MAPT, encoding tau, whose aggregation defines a family of tauopathies; TREM2, the microglial receptor that links neurodegeneration to the brain's immune compartment; and APOE, the shared bridge with Panel B. The Parkinson module contains SNCA, encoding alpha-synuclein, the aggregating protein of Lewy bodies; LRRK2, the kinase whose hyperactive variants are the most common genetic cause of Parkinson disease; PARK7, encoding DJ-1, an oxidative-stress sensor; PINK1, the mitophagy-initiating kinase; and GBA, the lysosomal glucocerebrosidase whose carrier status is among the strongest known Parkinson risk factors.

The motor-neuron and frontotemporal module contains SOD1, the first discovered ALS gene; TARDBP, encoding TDP-43, whose mislocalisation and aggregation is the pathological signature of most ALS and much frontotemporal dementia; FUS, another aggregating RNA-binding protein; C9orf72, whose hexanucleotide repeat expansion is the most common inherited cause of both ALS and frontotemporal dementia; GRN, encoding progranulin; OPTN, encoding optineurin, an autophagy receptor mutated in ALS and glaucoma; SQSTM1, encoding the autophagy adaptor p62; VCP, the AAA-ATPase at the centre of multisystem proteinopathy; and UBQLN2, a ubiquitin-shuttling quality-control factor mutated in ALS with dementia. The repeat and prion module contains HTT, whose polyglutamine expansion causes Huntington disease, and PRNP, the prion protein. Finally, the trophic module contains BDNF and NGF, the neurotrophins whose signalling sustains neuronal survival and plasticity and whose decline tracks cognitive aging.

Read against the panel, a pattern is already visible before any jellyfish data are considered: most of these genes are not neural-specific. They are maintenance genes, handling protein quality control, lysosomal clearance, mitochondrial surveillance, stress response, and membrane trafficking, that happen to fail first in neurons because neurons are long-lived, post-mitotic, and unable to dilute damage by dividing. This is the deep reason a rejuvenating animal is relevant to

neurodegeneration: neurodegeneration is, mechanistically, accelerated local aging of the least replaceable cells. The same hallmarks that organise aging research organise the molecular pathology of Alzheimer, Parkinson, and ALS, as the converging-pathways literature in neurodegeneration has argued for a decade.

The specific questions the jellyfish map can answer

For neurology, the expansion asks three questions of Panel C. Which of these human disease genes have mappable counterparts in the *T. dohrnii* genome at all? Of those, which are shared with the mortal cnidarians, and which, if any, show copy-number or sequence divergence in the immortal species? And does the set of neurodegeneration-relevant genes present in the jellyfish overlap the longevity set more than chance, which would be genomic evidence for the maintenance-machinery bridge between the two fields? The first question produces an evolutionary annotation of the human disease-gene list: which vulnerabilities are ancient and which are new. The second nominates candidate rejuvenation-associated divergence in neural-relevant genes. The third is the pre-registered H5 test, and its answer, positive or negative, disciplines how strongly the aging-neurology connection can be stated from this dataset.

What this does not claim

It is important to be blunt about the limits. *T. dohrnii* does not get Alzheimer disease. It has no amyloid cascade to model, no dopaminergic system to lose, no prion to propagate. The presence of a mappable homolog of APP or SNCA in a cnidarian genome says something about the evolutionary age of the protein's normal function, and nothing direct about the human disease, which involves derived features of the human protein, human brain architecture, and human lifespan. The value of the map is upstream of disease modelling: it tells researchers which disease genes sit on ancient, jellyfish-shared foundations, where those foundations might be interrogated in a regenerative whole-animal context, and which disease processes are human- or vertebrate-specific and must be studied in vertebrate systems. A map that knows its own limits is more useful than a model that pretends to be a brain.

With the two connection chapters in place, the report now turns to the machinery of the expansion itself: the panels frozen before data access, the genomes, the calibrated pipeline, the controls, and the statistical protocol that every result in the following chapters had to survive.

## Methods

### Design overview

The expansion follows a pre-registered, genome-first, calibration-first design. Three commitments shape every step. First, nothing about the panels, hypotheses, thresholds, or statistical tests was changed after outcome data were touched; the analysis plan was committed to the repository before the first query protein was downloaded, and its commit hash is the temporal proof. Second, the mapping pipeline is inherited unchanged from the closed phase, so that new results inherit old calibration rather than freshly tuned judgement. Third, every genome-wide claim is accompanied by a positive control that must reproduce prior calls and a negative control that must stay quiet, and both controls are reported regardless of outcome.

### Genomes

Five genome assemblies are analysed. Two are the available *Turritopsis dohrnii* assemblies: GCA\_027922465.2, a contig-level assembly, and GCA\_025167195.1, the scaffold-level assembly associated with the Oviedo group; treating both gives an internal replication of every *T. dohrnii* call across independently produced assemblies. The non-reverting comparators are *Turritopsis rubra* GCA\_039566895.2, a chromosome-level assembly new to this expansion and the phylogenetically closest available mortal contrast, and *Aurelia aurita* GCA\_004194415.1, the moon jelly carried over from the closed phase as a more distantly related medusozoan. *Clytia hemisphaerica* GCF\_902728285.1, a RefSeq-annotated hydrozoan genome, serves as the annotated calibration reference and as a source of cnidarian query proteins. All assemblies were downloaded from NCBI over FTP/HTTPS, and every file's SHA-256 checksum is recorded in the repository's data provenance file together with its URL. A third *T. dohrnii* accession cited by earlier work, GCA\_051903475.1, was inspected and found to expose only assembly reports and statistics, with no sequence files served; it is recorded as unavailable rather than silently dropped.

### Query proteins and their retrieval

For each gene in Panels B and C, query proteins were retrieved from NCBI through the E-utilities protein database, following the closed phase's retrieval protocol: an organism-scoped search with the gene symbol in all fields restricted to RefSeq, falling back to a gene-name field

search when the first query returns nothing, and selection of the longest RefSeq record among the top candidates, with an alphabetical accession tiebreak so the procedure is deterministic. Human queries come from *Homo sapiens*. Cnidarian queries come from four annotated species, *Clytia hemisphaerica*, *Hydra vulgaris*, *Acropora millepora*, and *Nematostella vectensis*, because cross-species query sets raise sensitivity for deep homologs, exactly as the closed phase demonstrated when *Clytia* queries produced the strongest jellyfish-genome matches. Every retrieved accession and sequence length is logged in a manifest; genes for which no cnidarian query exists are mapped with the human query alone, and that asymmetry is visible in the manifest rather than hidden.

The mapping pipeline, frozen

Mapping uses tBLASTn 2.17.0 against per-genome BLAST databases built with makeblastdb. Search parameters are the closed phase's, unchanged: e-value threshold 1e-5, at most twenty target sequences per query, tabular output with query and subject coordinates and lengths. Query coverage is computed per high-scoring pair as the covered fraction of the query length. Hits for the same gene across all query species are merged and clustered into loci: high-scoring pairs on the same contig within fifty kilobases are grouped, each locus carries the sum of its members' bit scores, and a locus is called strong when the summed score reaches one hundred bits, the threshold frozen by the closed phase. For each gene and genome the ledger records the number of loci, the number of strong loci, and the best locus with its identity, coverage, strand, and coordinates. A gene is called present in a genome when it has at least one strong locus; its copy signal is the count of strong loci. This is a homology-mapping method, not a gene-prediction method: it establishes that sequence recognisably homologous to the query exists at a locus, and it does not attempt to build exon models, which is a separate and harder problem deliberately left out of scope.

Controls

Two controls gate interpretation, as pre-registered. The positive control re-runs the closed phase's Panel A queries against the rebuilt databases and requires the presence calls of the closed phase's ledger to reproduce; any mismatch is treated as a pipeline defect and blocks interpretation of the new panels until resolved. The negative control shuffles every Panel B and C query protein with

a fixed seed, preserving length and overall composition while destroying homology, and runs the shuffled set against every genome through the identical pipeline. The false-positive rate is the fraction of shuffled queries producing a strong locus, and the pre-registered validity criterion is five percent or less in every genome. If the negative control fails, the run is invalid and only the control may be used to diagnose why; the outcome data may not be re-tuned.

### Statistical protocol

Hypotheses H1 through H6 were fixed in the preregistration. H1 and H2 test whether at least half of Panel B and Panel C, respectively, map in the *T. dohrnii* assemblies, using a one-sided binomial test against the negative-control false-positive rate, with a small-count correction so a zero observed control rate does not produce a degenerate test. H3 tests the copy-number contrast per gene between the immortal species and the mortal comparators with Fisher exact tests, Benjamini-Hochberg corrected across the panel; the pre-registered reading is that surviving contrasts nominate candidate expansion genes, while no survivors is an honest negative for the expansion hypothesis in this panel. H4 tests whether the presence map separates the rejuvenating species from the non-reverting cnidarians, using the difference between within-Turritopsis-dohrnii and cross-species mean Jaccard distance as the statistic and an exact permutation over genome labels for the null; the known caveat, stated in the preregistration and repeated here, is that the two *T. dohrnii* entries are assemblies of one species, so within-group similarity partly reflects assembly redundancy, and H4 is interpreted with that limitation. H5 tests whether the genes mapping in *T. dohrnii* from Panels B and C overlap more than expected under independence, by Fisher exact test over the union gene universe. H6 is the controls hypothesis above. Benjamini-Hochberg correction is applied within each pre-registered family at  $q$  below 0.05, and every test, significant or not, appears in the results. Exploratory analyses are labelled as such and never upgrade a pre-registered negative.

### Implementation and reproducibility

All code is in the repository: the panel definitions in `jellyfish/xpanel.py`, the statistics in `jellyfish/xstats.py` with unit tests for every function, the fetch, mapping, and analysis drivers in `experiments/`. The test suite, which included forty-one tests at the unification commit, grows with the expansion; every new module ships with tests, including synthetic-case tests for the statistics

and frozen-list tests for the panels. The full pipeline runs on a two-core, one-gigabyte sandbox, which is itself a small methodological point: none of this requires a cluster, and every number in the report can be regenerated from the repository, the recorded checksums, and public data.

#### Page-count methodology for this report

The standing format rule for this program is fifty or more pages of body text, with headings, references, appendices, and diagrams excluded from the count, and diagrams explicitly not counting toward the total. The count in this report is computed mechanically: the body prose (chapters one through twelve, excluding headings, figure captions, the reference list, and appendices) is assembled as a standalone document, typeset at twelve-point with one-and-a-half line spacing, rendered to PDF, and its page count read from the rendered file. The formula, the body word count, and the rendered page count are reported in the colophon so the compliance number is checkable rather than asserted.

### **The five genomes: a field guide for the reader**

Every number in the results chapters lives or dies by the quality of the genome it was measured in, so this short chapter profiles the five assemblies as instruments before they are used as evidence.

*Turritopsis dohrnii*, contig-level assembly GCA\_027922465.2. This is the newer and cleaner of the two immortal-jellyfish assemblies: 891 sequences spanning about 436 megabases, consistent with published estimates of the *T. dohrnii* genome size. For a non-model marine invertebrate this is respectable continuity; genes of ordinary size will usually sit complete on one contig. When a claim in this report matters most, this assembly is the primary source, and its companion serves as replication.

*Turritopsis dohrnii*, Oviedo scaffold assembly GCA\_025167195.1. The assembly associated with the group that produced the reference comparative-genomics study is much more fragmented: 74,835 scaffolds spanning about 400 megabases, an average scaffold of barely five kilobases. Fragmentation at that scale has direct consequences for translated mapping: genes are routinely split across scaffolds, alignments truncate at scaffold ends, and copy-number estimates are depressed. Everywhere the two *T. dohrnii* assemblies disagree, this asymmetry is the first

suspect, and the report says so where it occurs. Its independent provenance still makes it valuable: a locus found in both assemblies is unlikely to be an assembly artefact.

*Turritopsis rubra*, GCA\_039566895.2. The non-reverting congener and the expansion's new comparator is the best assembly in the set: 326 sequences spanning about 267 megabases, chromosome-level. The smaller, cleaner genome is a methodological gift and an analytical hazard at once. Its genes map more completely, so contrasts against *T. dohrnii* can reflect instrument quality rather than biology; the pre-registered statistics therefore lean on presence and coarse copy classes rather than fine copy counts. Biologically, *T. rubra* is the comparison the field cares about: same genus, same basic life cycle, no rejuvenation. Whatever distinguishes the immortal species should be visible against this background more clearly than against any other.

*Aurelia aurita*, GCA\_004194415.1. The moon jelly is the distant medusozoan anchor: 2,709 sequences spanning about 377 megabases, carried over from the closed phase. *Aurelia* is a scyphozoan, separated from the hydrozoan *Turritopsis* by hundreds of millions of years, so it cannot adjudicate the immortal-versus-mortal contrast within *Turritopsis*. Its role is calibration and breadth: a gene found in *Aurelia* and *Hydra* and *Clytia* is ordinary cnidarian machinery, and a gene pattern unique to *T. dohrnii* among these genomes earns a second look precisely because *Aurelia* stands for the medusozoan mainstream.

*Clytia hemisphaerica*, GCF\_902728285.1. The annotated hydrozoan reference: 1,396 sequences spanning about 421 megabases, with a full RefSeq annotation. *Clytia* is the calibration genome. Its annotated genes tell us what the mapping pipeline ought to find in a hydrozoan, its RefSeq proteins supply the most sensitive cross-species queries for the jellyfish genomes, and its annotated status makes it the one genome where the pipeline's calls can be checked against independent truth. When the controls chapter says the pipeline behaves, *Clytia* is where much of that confidence is earned.

One asymmetry deserves plain statement before the results. The immortal species is measured with the two most problematic instruments in the set: a decent contig assembly and a badly fragmented scaffold assembly, while its key comparator, *T. rubra*, enjoys the best. This stacks the deck against finding extra copies in the immortal species, because fragmentation hides them. Any copy-number advantage that survives that deck-stacking is thereby more interesting, and

any absence must be weighed against the possibility that fragmentation, not evolution, did the removing.

## **Working with an unannotated genome: a practical appendix in prose**

Because the annotation desert is the organising fact of this project, this chapter records, in plain procedural prose, what actually worked, so that the next project on an unannotated organism can skip the false starts. It is written as methods narrative rather than protocol, and it is deliberately specific about failures as well as successes.

The first lesson is that accession hygiene is half the battle. Genome accessions come in GenBank (GCA) and RefSeq (GCF) flavours, and the difference is not cosmetic: the GCF twin of an assembly is the one that may carry an annotation, while the GCA original may expose nothing but sequence, or, as with one *T. dohrnii* accession we inspected, nothing but reports. Before designing any analysis, enumerate the assembly hub for the exact accession and confirm that sequence files are actually served. Ten minutes of accession hygiene would have saved the field's naive reading of the database gap, and it saved this project from designing around a genome that was not there.

The second lesson is that query proteins are the sensitivity budget. A translated search is only as good as the proteins thrown at it, and the closed phase demonstrated the asymmetry concretely: human queries find the strongly conserved core, but cnidarian queries, being evolutionarily closer to the jellyfish target, find more and score higher. The expansion therefore never maps with a human query alone when a cnidarian homolog exists; the retrieval protocol walks four annotated cnidarian species per gene and keeps the longest RefSeq record per species, deterministically. The manifest of what was actually used is part of the result, because a gene absent under a human-only query and present under a cnidarian query is not a discrepancy; it is the expected behaviour of the method.

The third lesson is that thresholds must be inherited, not rediscovered. Every calibration choice in this pipeline, the e-value, the bit-score floor for a strong locus, the clustering span, was fixed by the closed phase before the expansion's data existed. The temptation in any new panel is to nudge a threshold when a beloved gene comes up absent; the preregistration forbids it, and the

honest-negative register is where the beloved absent gene is written down. This is not asceticism for its own sake. The entire credibility of a five-genome, sixty-nine-gene screen rests on the reader knowing the instrument was not re-tuned per gene.

The fourth lesson is that controls are the analysis, not an appendix to it. The positive control, re-running the old panel on the rebuilt databases, catches silent pipeline rot: a database version change, a truncated download, a path mistake, all announce themselves as failures to reproduce known loci before any new claim is built on them. The negative control, shuffled queries through the identical pipeline, prices the false-positive rate of the whole apparatus in each genome separately, and it priced differently across assemblies, which is itself information about the assemblies. Both controls run on the same hardware, in the same week, against the same database files as the real mapping. A control from another day is a control of another pipeline.

The fifth lesson is that small computers are a feature. The entire expansion, five genomes, three panels, controls, statistics, and this paper, runs on a two-core machine with one gigabyte of memory, which forces streaming designs, sequential jobs, and honest bookkeeping about what is in memory when. The result is a pipeline a student can rerun on a laptop, which matters because reproducibility that requires a cluster is reproducibility most of the world cannot check. The constraint also disciplines scope: when the run queue is hours long, pre-registration stops being a virtue and becomes a scheduling necessity, because nobody wants to discover, at hour three, that the wrong threshold was run.

The sixth lesson is about endings. Every run in this project terminates in a ledger and a checksum, not in a figure. Figures are for readers; ledgers are for checkers. The repository carries both, but the ledgers are the source of truth from which every figure, table, and sentence of this paper is generated, and the page you are reading exists because a script turned those ledgers into prose. That inversion, prose generated from checkable data rather than prose decorated with data, is the working style the annotation desert forces on you, and it is, in the end, the project's most exportable habit.

This section collects the ten numbered formulas used in the pipeline and analyses; each is referenced by number where it is used.

(1) Locus clustering. HSPs on one contig are sorted by genomic start and chained while the gap to the running locus end stays within the frozen span:  $\text{locus} = \{ h_i : s_{i+1} - \text{max\_end}(h_1..h_i) \leq 50 \text{ kb} \}$ . A cluster becomes one candidate locus with coordinates [min start, max end].

(2) Total locus bits.  $\text{bits}(\text{locus}) = \sum_i \text{bits}(h_i)$ , the sum of bitscores of all HSPs assigned to the locus.

(3) Strong-locus criterion. A locus is strong iff  $\text{bits}(\text{locus}) \geq 100$ . Presence of a gene in a genome is defined as at least one strong locus; the copy signal is the number of strong loci.

(4) Query coverage.  $\text{qcov}(\text{locus}) = \max \text{ over HSPs of } (q_{\text{end}} - q_{\text{start}} + 1) / q_{\text{len}}$ , the largest fraction of the query protein covered by a single HSP in the locus, used as the domain-integrity proxy.

(5) Shuffled-control false-positive rate.  $\text{FPR}(\text{genome}) = (\text{number of shuffled queries yielding at least one strong locus}) / (\text{number of shuffled queries})$ , with the pre-registered validity criterion  $\text{FPR} \leq 0.05$ .

(6) Binomial mapping test (H1, H2). For  $k$  mapped genes of  $n$  panel genes with control rate  $p_0$ ,  $p = P(X \geq k)$  for  $X \sim \text{Binomial}(n, p_0)$ , with  $p_0 = \max(\text{FPR}, 1/126)$ , the conservative upper bound from 125 shuffled queries.

(7) Benjamini-Hochberg adjustment. For  $m$  p-values sorted  $p_{(1)} \leq \dots \leq p_{(m)}$ ,  $q_{(i)} = \min \text{ over } j \geq i \text{ of } (m/j) * p_{(j)}$ , applied per hypothesis family at the frozen 0.05 level.

(8) Copy-share Fisher contrast (H3). For gene  $g$  with strong-locus counts  $c_{\text{Td}}$  and  $c_{\text{ot}}$  and group totals  $T_{\text{Td}}$ ,  $T_{\text{ot}}$ , the 2x2 table  $[[c_{\text{Td}}, c_{\text{ot}}], [T_{\text{Td}} - c_{\text{Td}}, T_{\text{ot}} - c_{\text{ot}}]]$  is tested by two-sided Fisher exact, with BH across genes.

(9) Jaccard distance and separation statistic (H4). For presence sets  $A, B$ :  $d_J(A, B) = 1 - |A \cap B| / |A \cup B|$ ; the separation statistic is mean  $d_J$  within the T. dohrnii assemblies

minus mean  $d_J$  between *T. dohrnii* and the non-reverting genomes, tested exactly over all 6 label assignments.

(10) Assembly-aware evolutionary evidence score.  $AEES = 0.35 * \min(\text{bits}/500, 1) + 0.25 * \text{qcov} + 0.15 * \text{pident} + 0.25 * \text{concordance}$ , where concordance in  $\{0, 0.5, 1\}$  counts detection in neither, one, or both *T. dohrnii* assemblies; for single-assembly genomes the first three weights renormalize by  $1/0.75$ . Confidence bands: high  $\geq 0.75$ , moderate  $\geq 0.45$ , low below.

(11) Recall (benchmark arms).  $\text{recall} = (\text{genes with at least one strong locus under the arm}) / (\text{panel size})$ , compared paired: multi-species versus human-only queries on *Clytia*, and genome-first versus database-first on *T. dohrnii*.

Formulas 1-3 and 5-6 are pre-registered and frozen; formula 8 implements the pre-registered Fisher contrast and its exact table is documented in docs/X-DEVIATIONS.md (D2); formula 10 is the judge-round R1 addition and is exploratory; formula 11 is the benchmark measurement. The numbering is used throughout the results chapters.

## The Genome-First Annotation Recovery workflow, stated formally

Judge round one asked whether the method is anything more than BLAST applied carefully. The answer, now formalized in full in docs/X-ALGORITHM.md, is that the contribution is the workflow around the aligner: a named pipeline (GFAR) with fixed inputs, frozen parameters, an evidence score, and a validation battery, designed for genomes with no annotation of any kind. Its inputs are a raw assembly, a gene panel whose every query is exact-gene verified against its source database, a cross-species query expansion set that shrinks query-to-target distance along several independent evolutionary paths, and comparator assemblies for concordance. Its stages are panel verification, *tblastn* mapping at e-value at most  $1e-10$ , locus clustering (HSPs of at least 100 bits merged across gaps of at most 500 base pairs), strong-locus and significance calls (a gene is significant with at least 3 strong loci), two-assembly concordance, and scoring by the assembly-aware evidence score  $AEES = 0.35 \times \text{normalized bitscore} + 0.25 \times \text{query coverage} + 0.15 \times \text{percent identity} + 0.25 \times \text{assembly concordance}$ , with weights frozen before outcomes were read. Every claim must additionally pass a validation battery: shuffled-query false-positive estimation, a verified non-aging decoy panel, reciprocal-best-hit orthology,  $dN/dS$ , orthogonal

search engines, and a micro-synteny scan whose expected desert negative is reported rather than suppressed. The workflow is benchmarked as a workflow (item ten of the judge round): against human-query-only mapping it recovers 35 of 47 multi-copy panel genes versus 27 of 47, and against the standard desert route of untrained de novo prediction it recovers 35 of 47 versus 0 of 47. It runs on two cores and two gigabytes of memory for half-gigabase assemblies, which is the point: annotation recovery for the thousands of neglected genomes that will never receive a model-organism investment.

## Results: the aging and longevity panel, gene by gene

Panel headline (pre-registered H1): 20 of 24 Panel B genes map in at least one *T. dohrnii* assembly (fraction 0.83); binomial  $p = 1.01e-38$  against the negative-control rate 0.00794, Benjamini-Hochberg  $q = 2.03e-38$  within the H1-H2 family. Present: AKT1, ATG5, ATG7, ATM, ATR, BECN1, BLM, FOXO3, HSF1, IGF1R, INSR, MTOR, NFE2L2, PRKAA1, PTEN, SIRT1, SIRT6, TFEB, TP53, WRN. Absent: APOE, CDKN2A, GHR, IGF1.

FOXO3 (stress/longevity-TF). FOXO3 is the most consistently replicated longevity-associated gene in human populations; protective alleles enrich in centenarians across ancestries. As a forkhead transcription factor it integrates insulin/IGF signalling with stress resistance, upregulating antioxidant, autophagy, and repair programmes when growth signalling is low. Its homologs extend lifespan in worms and flies, making it the bridge gene between the invertebrate and human longevity literatures. MAPPED: FOXO3 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 2 strong locus/i; best locus on contig dbj|BQMF02000104.1|:714816-716011, 2 high-scoring pairs, total bit score 196.2, peak identity 79.661%, peak query coverage 26%. In *T. dohrnii* (Oviedo scaffold assembly): 2 strong locus/i; best locus on contig gb|JAIFHF010001213.1|:20794-21989, 2 high-scoring pairs, total bit score 196.2, peak identity 79.661%, peak query coverage 26%. In *T. rubra*: 2 strong locus/i; best locus on contig gb|CM077624.1|:5891950-5893233, 2 high-scoring pairs, total bit score 197.0, peak identity 70.588%, peak query coverage 29%. In *A. aurita*: 2 strong locus/i; best locus on contig gb|REGM01000007.1|:273714-275029, 2 high-scoring pairs, total bit score 193.7, peak identity 80.0%, peak query coverage 20%. Comparator status: present in *T. rubra*, present in *A. aurita*.

SIRT1 (sirtuin). SIRT1 is an NAD-dependent deacetylase that couples metabolic state to chromatin and transcription. It deacetylates p53, FOXO factors, and PGC-1 $\alpha$ , and its overexpression improves metabolic health in mammals, though lifespan effects are context-dependent. It is the most studied sirtuin and the canonical link between calorie restriction and gene regulation. MAPPED: SIRT1 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 6 strong locus/i; best locus on contig dbj|BQMF02000020.1|:841937-845794, 23 high-scoring pairs, total bit score 1820.2, peak identity 94.118%, peak query coverage 20%. In *T. dohrnii* (Oviedo scaffold assembly): 5 strong locus/i; best locus on contig gb|JAIFHF010000084.1|:46180-49837, 20 high-scoring pairs, total bit score 1707.4, peak identity 94.118%, peak query coverage 20%. In *T. rubra*: 5 strong locus/i; best locus on contig gb|CM077625.1|:6489093-6495329, 17 high-scoring pairs, total bit score 1691.7, peak identity 88.462%, peak query coverage 27%. In *A. aurita*: 4 strong locus/i; best locus on contig gb|REGM01000101.1|:943978-948809, 13 high-scoring pairs, total bit score 890.9, peak identity 92.593%, peak query coverage 13%. Comparator status: present in *T. rubra*, present in *A. aurita*.

SIRT6 (sirtuin). SIRT6 is the sirtuin most directly tied to genome maintenance: it promotes double-strand-break repair, stabilises telomeric chromatin, and suppresses LINE1 retrotransposons. Sirt6-null mice age prematurely, and overexpression extends lifespan in males of some backgrounds. Among the panel it is the cleanest single-protein argument that repair quality sets aging pace. MAPPED: SIRT6 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 4 strong locus/i; best locus on contig dbj|BQMF02000027.1|:677976-682990, 20 high-scoring pairs, total bit score 1487.7, peak identity 88.095%, peak query coverage 25%. In *T. dohrnii* (Oviedo scaffold assembly): 3 strong locus/i; best locus on contig gb|JAIFHF010003447.1|:8223-11303, 20 high-scoring pairs, total bit score 1496.9, peak identity 88.095%, peak query coverage 25%. In *T. rubra*: 3 strong locus/i; best locus on contig gb|CM077619.1|:7495336-7501012, 21 high-scoring pairs, total bit score 1564.1, peak identity 88.0%, peak query coverage 29%. In *A. aurita*: 3 strong locus/i; best locus on contig gb|REGM01000159.1|:323452-327589, 14 high-scoring pairs, total bit score 1235.3, peak identity 71.429%, peak query coverage 33%. Comparator status: present in *T. rubra*, present in *A. aurita*.

MTOR (nutrient-sensing). The mTOR kinase is the central growth regulator of eukaryotes and the most pharmacologically validated aging target: rapamycin extends lifespan in yeast, worms,

flies, and mice, the only drug-class effect of that breadth. MTOR integrates amino-acid, energy, and growth-factor inputs; its chronic activation drives the growth-first, maintenance-second physiology that aging theory blames for late-life decline. MAPPED: MTOR is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 6 strong locus/i; best locus on contig dbj|BQMF02000020.1|:293731-311627, 107 high-scoring pairs, total bit score 9544.9, peak identity 93.103%, peak query coverage 8%. In *T. dohrnii* (Oviedo scaffold assembly): 12 strong locus/i; best locus on contig gb|JAIFHF010017744.1|:71-5685, 43 high-scoring pairs, total bit score 4438.4, peak identity 93.478%, peak query coverage 5%. In *T. rubra*: 7 strong locus/i; best locus on contig gb|CM077625.1|:7792069-7812162, 77 high-scoring pairs, total bit score 8601.2, peak identity 91.379%, peak query coverage 14%. In *A. aurita*: 6 strong locus/i; best locus on contig gb|REGM01000013.1|:6978-82123, 99 high-scoring pairs, total bit score 7646.4, peak identity 100.0%, peak query coverage 4%. Comparator status: present in *T. rubra*, present in *A. aurita*.

IGF1 (insulin/IGF). Insulin-like growth factor 1 is the effector of the growth-hormone axis. Circulating IGF1 declines with age, but reduced IGF1 signalling extends lifespan in model organisms, a paradox resolved by distinguishing development-time growth promotion from adult maintenance suppression. Small, long-lived dog breeds and long-lived human kindreds sit at the low-signalling end. NOT MAPPED: no strong locus for IGF1 in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

IGF1R (insulin/IGF). The IGF1 receptor transduces the IGF signal. Haploinsufficiency extends female lifespan in mice, and low IGF1R activity marks human longevity kindreds. It is one of the few receptors whose partial inhibition is life-extending, which makes it a reference point for the difference between blocking a pathway and tuning it. MAPPED: IGF1R is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 24 strong locus/i; best locus on contig dbj|BQMF02000012.1|:1834423-1842309, 4 high-scoring pairs, total bit score 479.4, peak identity 65.217%, peak query coverage 10%. In *T. dohrnii* (Oviedo scaffold assembly): 19 strong locus/i; best locus on contig gb|JAIFHF010006514.1|:6423-8020, 3 high-scoring pairs, total bit score 413.0, peak identity 65.217%, peak query coverage 10%. In *T. rubra*: 33 strong locus/i; best locus on contig gb|CM077619.1|:5195038-5206233, 5 high-scoring pairs, total bit score 512.3, peak identity 86.364%, peak query coverage 8%. In *A. aurita*: 19 strong locus/i; best locus on contig gb|REGM01000087.1|:194367-257861, 9 high-scoring pairs, total bit score 1064.7, peak

identity 81.72%, peak query coverage 11%. Comparator status: present in *T. rubra*, present in *A. aurita*.

INSR (insulin/IGF). The insulin receptor is the entry point of the insulin signalling cascade. Its worm homolog daf-2 is the founding longevity gene of the modern field: reducing it doubles worm lifespan. INSR sits at the top of the most evolutionary conserved aging pathway known, which is why its handling in an ancient metazoan is intrinsically informative. MAPPED: INSR is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 32 strong locus/i; best locus on contig dbj|BQMF02000012.1|:1834012-1842366, 40 high-scoring pairs, total bit score 4296.9, peak identity 86.486%, peak query coverage 16%. In *T. dohrnii* (Oviedo scaffold assembly): 23 strong locus/i; best locus on contig gb|JAIFHF010006514.1|:6012-12288, 32 high-scoring pairs, total bit score 3521.1, peak identity 86.486%, peak query coverage 16%. In *T. rubra*: 71 strong locus/i; best locus on contig gb|CM077619.1|:5194627-5206359, 48 high-scoring pairs, total bit score 4630.7, peak identity 90.909%, peak query coverage 16%. In *A. aurita*: 26 strong locus/i; best locus on contig gb|REGM01000087.1|:194337-257939, 60 high-scoring pairs, total bit score 6934.6, peak identity 85.859%, peak query coverage 14%. Comparator status: present in *T. rubra*, present in *A. aurita*.

AKT1 (insulin/IGF). AKT1 is the central kinase of insulin/IGF signalling, phosphorylating FOXO factors to exclude them from the nucleus and activating mTOR signalling. In longevity terms it is the switch that decides between growth and maintenance. Its deep conservation makes it a near-certain find in any metazoan genome and a calibration point for the panel. MAPPED: AKT1 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 20 strong locus/i; best locus on contig dbj|BQMF02000318.1|:310713-348513, 12 high-scoring pairs, total bit score 759.7, peak identity 81.818%, peak query coverage 17%. In *T. dohrnii* (Oviedo scaffold assembly): 18 strong locus/i; best locus on contig gb|JAIFHF010000426.1|:27647-32674, 6 high-scoring pairs, total bit score 376.7, peak identity 94.286%, peak query coverage 14%. In *T. rubra*: 25 strong locus/i; best locus on contig gb|CM077621.1|:11357763-11415532, 23 high-scoring pairs, total bit score 1410.9, peak identity 85.714%, peak query coverage 18%. In *A. aurita*: 12 strong locus/i; best locus on contig gb|REGM01000167.1|:243174-254921, 6 high-scoring pairs, total bit score 455.4, peak identity 81.818%, peak query coverage 12%. Comparator status: present in *T. rubra*, present in *A. aurita*.

PTEN (insulin/IGF). PTEN is the lipid phosphatase that opposes PI3K signalling, restraining the AKT-mTOR axis, and a major tumour suppressor. Subtle PTEN elevation extends mouse lifespan with improved metabolic health. It exemplifies the class of maintenance genes whose loss is catastrophic and whose mild gain is protective. MAPPED: PTEN is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 3 strong locus/i; best locus on contig dbj|BQMF02000053.1|:678795-706481, 22 high-scoring pairs, total bit score 1722.8, peak identity 81.034%, peak query coverage 8%. In *T. dohrnii* (Oviedo scaffold assembly): 6 strong locus/i; best locus on contig gb|JAIFHF010006623.1|:8204-10355, 15 high-scoring pairs, total bit score 1193.2, peak identity 81.034%, peak query coverage 6%. In *T. rubra*: 3 strong locus/i; best locus on contig gb|CM077621.1|:16699033-16731511, 23 high-scoring pairs, total bit score 1747.1, peak identity 70.0%, peak query coverage 10%. In *A. aurita*: 4 strong locus/i; best locus on contig gb|REGM01000023.1|:400814-406232, 11 high-scoring pairs, total bit score 873.7, peak identity 76.667%, peak query coverage 5%. Comparator status: present in *T. rubra*, present in *A. aurita*.

TP53 (genome-guardian). p53 is the genome guardian: it senses DNA damage and forces arrest, senescence, or apoptosis. Aging biology's double bind is concentrated here: too little p53 activity yields cancer, too much yields stem-cell attrition and progeroid phenotypes. Cnidarian p53 homologs are ancient, and how a non-aging animal sets the p53 dial is a genuine open question. MAPPED: TP53 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 3 strong locus/i; best locus on contig dbj|BQMF02000243.1|:105726-119709, 3 high-scoring pairs, total bit score 169.8, peak identity 39.394%, peak query coverage 30%. In *T. dohrnii* (Oviedo scaffold assembly): 2 strong locus/i; best locus on contig gb|JAIFHF010000175.1|:42369-43465, 2 high-scoring pairs, total bit score 108.2, peak identity 38.636%, peak query coverage 26%. In *T. rubra*: 2 strong locus/i; best locus on contig gb|CM077631.1|:3377820-3378930, 2 high-scoring pairs, total bit score 125.9, peak identity 43.0%, peak query coverage 30%. Comparator status: present in *T. rubra*, absent in *A. aurita*.

CDKN2A (senescence). CDKN2A encodes p16INK4a, the senescence brake whose expression rises steeply with age and serves as a standard biomarker of senescent cell burden. Clearing p16-positive cells extends healthspan in mice. Its evolutionary distribution is uneven: it is a relatively recent vertebrate innovation, so its absence from a cnidarian genome would be the expected,

informative result. NOT MAPPED: no strong locus for CDKN2A in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

ATM (repair-signaling). ATM is the apical kinase of the double-strand-break response; its loss causes ataxia-telangiectasia, with neurodegeneration, immunodeficiency, and cancer predisposition. ATM links DNA repair directly to neurology: the most radiosensitive human tissue phenotype includes the cerebellum. It is ancient and nearly universal in eukaryotes.

MAPPED: ATM is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 5 strong locus/i; best locus on contig dbj|BQMF02000058.1|:669089-678923, 39 high-scoring pairs, total bit score 3223.2, peak identity 82.258%, peak query coverage 4%. In *T. dohrnii* (Oviedo scaffold assembly): 5 strong locus/i; best locus on contig gb|

JAIFHF010000980.1|:7816-16146, 35 high-scoring pairs, total bit score 3166.3, peak identity 70.0%, peak query coverage 4%. In *T. rubra*: 4 strong locus/i; best locus on contig gb|

CM077625.1|:10383669-10395304, 44 high-scoring pairs, total bit score 3405.0, peak identity 69.811%, peak query coverage 5%. In *A. aurita*: 4 strong locus/i; best locus on contig gb|

REGM01000023.1|:149484-208003, 57 high-scoring pairs, total bit score 4424.8, peak identity 87.5%, peak query coverage 4%. Comparator status: present in *T. rubra*, present in *A. aurita*.

ATR (repair-signaling). ATR handles replication stress and single-stranded DNA, and its hypomorphic loss causes Seckel syndrome with microcephaly and progeroid features. Together with ATM it organises the checkpoint layer of genome maintenance. Its essentiality makes presence in any metazoan genome expected and its divergence pattern the informative variable.

MAPPED: ATR is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 4 strong locus/i; best locus on contig dbj|BQMF02000094.1|:416321-427248, 59 high-scoring pairs, total bit score 7618.2, peak identity 84.906%, peak query coverage 17%. In *T. dohrnii* (Oviedo scaffold assembly): 5 strong locus/i; best locus on contig gb|

JAIFHF010002096.1|:12036-22788, 63 high-scoring pairs, total bit score 7301.7, peak identity 81.081%, peak query coverage 14%. In *T. rubra*: 6 strong locus/i; best locus on contig gb|

CM077623.1|:8374927-8391986, 94 high-scoring pairs, total bit score 8508.2, peak identity 84.314%, peak query coverage 10%. In *A. aurita*: 3 strong locus/i; best locus on contig gb|

REGM01000050.1|:844864-887094, 23 high-scoring pairs, total bit score 1872.6, peak identity 84.932%, peak query coverage 4%. Comparator status: present in *T. rubra*, present in *A. aurita*.

BLM (helicase). BLM is a RecQ helicase that dissolves recombination intermediates; biallelic loss causes Bloom syndrome, with genome instability, cancer predisposition, and growth failure. It is direct human genetic proof that helicase-based genome maintenance limits healthspan, and one of the clearest progeroid pathway genes. MAPPED: BLM is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 26 strong locus/i; best locus on contig dbj|BQMF02000234.1|:119680-125207, 23 high-scoring pairs, total bit score 2417.5, peak identity 90.476%, peak query coverage 14%. In *T. dohrnii* (Oviedo scaffold assembly): 19 strong locus/i; best locus on contig gb|JAIFHF010002901.1|:562-3726, 19 high-scoring pairs, total bit score 1856.9, peak identity 90.476%, peak query coverage 8%. In *T. rubra*: 64 strong locus/i; best locus on contig gb|CM077619.1|:6206626-6212170, 24 high-scoring pairs, total bit score 2292.7, peak identity 89.062%, peak query coverage 7%. In *A. aurita*: 24 strong locus/i; best locus on contig gb|REGM01000014.1|:1259074-1287706, 22 high-scoring pairs, total bit score 2050.6, peak identity 88.71%, peak query coverage 9%. Comparator status: present in *T. rubra*, present in *A. aurita*.

WRN (helicase). WRN is the RecQ helicase whose loss causes Werner syndrome, the canonical adult progeria: patients develop graying, cataracts, diabetes, atherosclerosis, and cancer by middle age. WRN is the strongest single-gene argument that genome-maintenance failure can phenocopy normal aging compressed in time, and its evolutionary history is notably patchy across animals. MAPPED: WRN is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 38 strong locus/i; best locus on contig dbj|BQMF02000105.1|:762247-931817, 25 high-scoring pairs, total bit score 1566.2, peak identity 27.941%, peak query coverage 25%. In *T. dohrnii* (Oviedo scaffold assembly): 35 strong locus/i; best locus on contig gb|JAIFHF010013166.1|:5463-6991, 6 high-scoring pairs, total bit score 953.9, peak identity 70.27%, peak query coverage 19%. In *T. rubra*: 72 strong locus/i; best locus on contig gb|CM077625.1|:4710952-4725907, 12 high-scoring pairs, total bit score 1186.2, peak identity 79.365%, peak query coverage 19%. In *A. aurita*: 58 strong locus/i; best locus on contig gb|REGM01000401.1|:12248-169099, 59 high-scoring pairs, total bit score 4059.9, peak identity 29.932%, peak query coverage 29%. Comparator status: present in *T. rubra*, present in *A. aurita*.

NFE2L2 (stress/NRF2). NFE2L2 encodes NRF2, the master transcription factor of the antioxidant and electrophile response. NRF2 activity declines with age, its activation extends

lifespan in models, and its pathway is the target of several approved and candidate drugs. In regeneration-competent animals the redox environment is actively managed; NRF2 is the likeliest transcription factor behind that management. MAPPED: NFE2L2 is present in the contig assembly only. In *T. dohrnii* (contig assembly): 1 strong locus/i; best locus on contig dbj|BQMF02000013.1|:1741013-1751056, 2 high-scoring pairs, total bit score 161.0, peak identity 39.394%, peak query coverage 22%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077620.1|:5862737-5871598, 2 high-scoring pairs, total bit score 157.1, peak identity 39.844%, peak query coverage 21%. Comparator status: present in *T. rubra*, absent in *A. aurita*.

HSF1 (proteostasis). HSF1 is the heat-shock transcription factor, guardian of the proteome under stress. Its decline with age tracks the collapse of proteostasis, and its activation extends worm lifespan. It anchors the panel's proteostasis arm alongside autophagy, which matters doubly for the neurology connection because aggregation diseases are proteostasis failures. MAPPED: HSF1 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 2 strong locus/i; best locus on contig dbj|BQMF02000075.1|:188814-189203, 2 high-scoring pairs, total bit score 173.7, peak identity 47.748%, peak query coverage 19%. In *T. dohrnii* (Oviedo scaffold assembly): 2 strong locus/i; best locus on contig gb|JAIFHF010001069.1|:14106-14495, 2 high-scoring pairs, total bit score 173.7, peak identity 47.748%, peak query coverage 19%. In *T. rubra*: 4 strong locus/i; best locus on contig gb|CM077629.1|:10939070-10971966, 6 high-scoring pairs, total bit score 355.4, peak identity 68.571%, peak query coverage 10%. In *A. aurita*: 3 strong locus/i; best locus on contig gb|REGM01000020.1|:331976-332311, 2 high-scoring pairs, total bit score 121.7, peak identity 37.037%, peak query coverage 20%. Comparator status: present in *T. rubra*, present in *A. aurita*.

ATG5 (autophagy). ATG5 is core autophagy machinery, conjugating ATG12 in the pathway that builds autophagosomes. Neural-specific Atg5 loss causes neurodegeneration in mice without any other trigger, making it a direct genetic link between maintenance failure and neuronal death. It is deeply conserved and expected in any eukaryote. MAPPED: ATG5 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 3 strong locus/i; best locus on contig dbj|BQMF02000106.1|:66386-66652, 3 high-scoring pairs, total bit score 455.0, peak identity 80.233%, peak query coverage 68%. In *T. dohrnii* (Oviedo scaffold assembly): 3 strong locus/i; best locus on contig gb|JAIFHF010011888.1|:5579-5845, 3 high-scoring pairs, total bit score

452.0, peak identity 80.233%, peak query coverage 68%. In *T. rubra*: 2 strong locus/i; best locus on contig gb|CM077619.1|:10947786-10948052, 3 high-scoring pairs, total bit score 453.0, peak identity 80.233%, peak query coverage 68%. In *A. aurita*: 2 strong locus/i; best locus on contig gb|REGM01000194.1|:502144-518495, 4 high-scoring pairs, total bit score 228.8, peak identity 68.75%, peak query coverage 24%. Comparator status: present in *T. rubra*, present in *A. aurita*.

ATG7 (autophagy). ATG7 is the E1 enzyme activating both ATG12 and LC3 in autophagy. Its restoration in old mice improves healthspan, and its loss phenocopies aspects of accelerated aging. Like ATG5 it is expected to be universal in eukaryotes, so its copy number and divergence, not its presence, are the informative readouts. MAPPED: ATG7 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 1 strong locus/i; best locus on contig dbj|BQMF02000060.1|:1053587-1069048, 28 high-scoring pairs, total bit score 2981.2, peak identity 92.308%, peak query coverage 32%. In *T. dohrnii* (Oviedo scaffold assembly): 2 strong locus/i; best locus on contig gb|JAIFHF010001824.1|:17520-23436, 25 high-scoring pairs, total bit score 2801.0, peak identity 92.308%, peak query coverage 37%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077625.1|:3437362-3446661, 44 high-scoring pairs, total bit score 3040.6, peak identity 89.189%, peak query coverage 15%. In *A. aurita*: 2 strong locus/i; best locus on contig gb|REGM01000001.1|:1601867-1606059, 13 high-scoring pairs, total bit score 996.6, peak identity 83.673%, peak query coverage 13%. Comparator status: present in *T. rubra*, present in *A. aurita*.

BECN1 (autophagy). Beclin-1 initiates autophagosome formation and connects autophagy to endosomal trafficking. A gain-of-function beclin variant extends mouse lifespan with reduced cardiac and renal pathology. BECN1 is also haploinsufficient-tumour-suppressive, placing it in the maintenance-genes-that-are-also-cancer-genes class that dominates the aging-cancer interface. MAPPED: BECN1 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 2 strong locus/i; best locus on contig dbj|BQMF02000011.1|:2000754-2004207, 10 high-scoring pairs, total bit score 846.6, peak identity 73.874%, peak query coverage 21%. In *T. dohrnii* (Oviedo scaffold assembly): 3 strong locus/i; best locus on contig gb|JAIFHF010012131.1|:4038-7491, 10 high-scoring pairs, total bit score 849.3, peak identity 73.874%, peak query coverage 21%. In *T. rubra*: 2 strong locus/i; best locus on contig gb|CM077623.1|:369940-377845, 12 high-scoring pairs, total bit score 915.9, peak identity

78.689%, peak query coverage 17%. In *A. aurita*: 2 strong locus/i; best locus on contig gb|REGM01000118.1|:550667-555480, 4 high-scoring pairs, total bit score 353.3, peak identity 54.237%, peak query coverage 22%. Comparator status: present in *T. rubra*, present in *A. aurita*.

TFEB (lysosome). TFEB is the master regulator of lysosomal biogenesis and autophagic flux, shuttling to the nucleus when nutrients are scarce. Its overexpression clears aggregation-prone proteins in models of Alzheimer, Parkinson, and lysosomal storage disease, making it the panel's most direct link from longevity machinery to neurodegeneration therapy research. MAPPED: TFEB is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 2 strong locus/i; best locus on contig dbj|BQMF02000058.1|:1281058-1283881, 18 high-scoring pairs, total bit score 907.5, peak identity 92.0%, peak query coverage 25%. In *T. dohrnii* (Oviedo scaffold assembly): 1 strong locus/i; best locus on contig gb|JAIFHF010001882.1|:7285-12930, 19 high-scoring pairs, total bit score 960.3, peak identity 92.0%, peak query coverage 25%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077625.1|:3974681-3977697, 18 high-scoring pairs, total bit score 893.8, peak identity 96.0%, peak query coverage 32%. In *A. aurita*: 1 strong locus/i; best locus on contig gb|REGM01000030.1|:2377020-2389144, 2 high-scoring pairs, total bit score 109.0, peak identity 40.217%, peak query coverage 21%. Comparator status: present in *T. rubra*, present in *A. aurita*.

PRKAA1 (AMPK). PRKAA1 encodes the catalytic alpha1 subunit of AMPK, the cell's energy-depletion sensor. AMPK activation extends lifespan in worms and flies and is implicated in the effects of metformin and exercise. It is the metabolic complement of mTOR: where mTOR drives growth, AMPK enforces maintenance during scarcity. MAPPED: PRKAA1 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 19 strong locus/i; best locus on contig dbj|BQMF02000057.1|:1067096-1068644, 5 high-scoring pairs, total bit score 337.8, peak identity 84.615%, peak query coverage 11%. In *T. dohrnii* (Oviedo scaffold assembly): 16 strong locus/i; best locus on contig gb|JAIFHF010000874.1|:18686-21937, 2 high-scoring pairs, total bit score 312.0, peak identity 37.313%, peak query coverage 46%. In *T. rubra*: 23 strong locus/i; best locus on contig gb|CM077618.1|:10862988-10918053, 17 high-scoring pairs, total bit score 1117.9, peak identity 84.444%, peak query coverage 10%. In *A. aurita*: 15 strong locus/i; best locus on contig gb|REGM01000038.1|:565059-572810, 4 high-scoring pairs, total bit score

281.6, peak identity 72.131%, peak query coverage 10%. Comparator status: present in *T. rubra*, present in *A. aurita*.

GHR (GH axis). The growth hormone receptor anchors the somatotrophic axis. Its inactivation produces Laron syndrome in humans, with marked protection from cancer and diabetes, and the longest-lived mouse on record is a GH-pathway mutant. GHR is a vertebrate-specific gene, so its expected absence from cnidarian genomes marks a boundary of the ancient maintenance toolkit. NOT MAPPED: no strong locus for GHR in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

APOE (lipid/neuro). Apolipoprotein E is the strongest common genetic factor for both human longevity and Alzheimer disease: the e4 allele raises Alzheimer risk and shortens lifespan, e2 does the reverse. APOE is a vertebrate lipid-transport gene, expected to be absent from cnidarians; its role in this study is to mark, by contrast, which parts of the aging-neurology interface are evolutionary novelties. NOT MAPPED: no strong locus for APOE in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

The assembly-aware evolutionary evidence score (AEES) was introduced to convert raw similarity calls into confidence-graded evolutionary evidence (method added in response to adversarial review, logged in docs/X-JUDGE-ROUNDS.md round R1; the layer is additive and does not alter the pre-registered presence and copy calls). For each gene in each genome, AEES combines four components with weights frozen before outcomes were read: the normalized total bitscore of the best locus (weight 0.35), the query coverage of that locus as a domain-integrity proxy (0.25), the best percent identity (0.15), and, for the two *T. dohrnii* assemblies only, a concordance term that scores 1.0 when a gene is detected in both assemblies, 0.5 in one, and 0.0 in neither (0.25). For single-assembly genomes the concordance weight redistributes proportionally over the remaining components. Genes with no strong locus are labeled not detected rather than absent, because a missing hit on a draft assembly cannot distinguish true loss from an assembly gap.

Applying the frozen high-confidence cutoff (AEES  $\geq 0.75$ ) yields 21 high-confidence genes in the main *T. dohrnii* assembly and 20 in the fragmented Oviedo assembly, against 4 in *T. rubra* and 1 in *A. aurita*, with the annotated *Clytia* calibration at 10. The top of the high-confidence *T. dohrnii* set, with scores, is: SOD1 (0.86), ATG5 (0.86), APP (0.83), ATG7 (0.82), PSEN1 (0.82), GRN (0.80), TFEB (0.80), SIRT6 (0.79), SIRT1 (0.79), VCP (0.78), TARDBP (0.78), HTT (0.78). GHR, which appeared in this list before the D9 query-integrity repair, is absent from every study genome under its verified human query and has been removed from the candidate set; SOD1 and TFEB entered the top ranks only after the repair, because their original human query files had contained mislabeled proteins. The contrast between the two mortal relatives (2-3 high-confidence genes) and the two immortal-lineage assemblies (17-20) is descriptive at this stage; the formal tests are H1-H4, and the per-gene scores are reported in full in results/x-aees.json. The concordance component also quantifies the assembly-quality problem directly: genes detected in only one *T. dohrnii* assembly carry concordance 0.5 and visibly lower scores, which is exactly the uncertainty the second assembly was kept to measure.

AEES serves two functions in this project. Internally, it is the confidence rubric for every downstream claim: any gene discussed as a candidate carries its score and its concordance status. Externally, it is a methodological contribution for comparative genomics on fragmented non-model genomes, where the standard output (hit found, hit not found) throws away exactly the evidence gradient that distinguishes a solid call from an assembly artifact. The score's limits are stated plainly: it has no expression component because no *T. dohrnii* transcriptome was used in the frozen design, and its phylogenetic component is reduced to percent identity pending a tree-based extension; both extensions are registered as future arms rather than claimed here.

### External aging-database annotation of the 48-gene panel

To place the panel in the landscape of existing aging and longevity research, we annotated every panel gene against seven external resources in live queries executed for this expansion (registry and raw outputs: docs/X-TOOLS.md, results/xext/). The results quantify something the panel was designed to do: it is not an arbitrary gene list but a dense intersection of the strongest aging-relevant annotations available.

GenAge, the reference database of genes associated with aging in humans, contains 29 of the 48 panel genes (60.4%), including AKT1, APOE, APP, ATM, ATR, BDNF, BLM, CDKN2A, FOXO3, GHR, GRN, HSF1, HTT, IGF1, IGF1R, INSR, MAPT, MTOR, NFE2L2, NGF, PSEN1, PTEN, SIRT1, SIRT6, SOD1, SQSTM1, TP53, VCP and WRN. The GenAge model-organism compendium (curated lifespan-modifying genes in yeast, worm, fly and mouse) contains 20 panel genes, among them ATG5 and ATG7 - direct experimental evidence that perturbing these autophagy genes changes lifespan in multiple model systems. CellAge, the database of genes driving cellular senescence, contains 21 panel genes including ATG5, ATG7 and BECN1, the three autophagy-initiation genes at the center of our discovery candidate. LongevityMap, the registry of human genetic association studies of longevity, contains 22 panel genes.

Pathway-level annotation agrees. The KEGG longevity-regulating pathways (hsa04211 and hsa04213) contain seven panel genes: FOXO3, IGF1, IGF1R, INSR, MTOR, PRKAA1 and SIRT1 - the insulin/IGF-1-mTOR-FOXO axis that is the most replicated longevity module in biology. The KEGG autophagy pathway (hsa04140) contains seven more: ATG7, C9orf72, IGF1R, MTOR, OPTN, PRKAA1 and SQSTM1. All 47 unique panel symbols resolve to a reviewed human UniProt entry (47/47).

The mirror image of this density is the jellyfish literature desert. Querying PubMed for "Turritopsis dohrnii" returns 14 publications in total, and intersecting the species with ATG5, TERT or SIRT1 returns zero papers each (results/xext/x-ext-pubmed-literature.json). Europe PMC, with broader indexing, returns 60 publications for the species but only 1 for ATG5, 1 for ATG7 and 0 for BECN1 (results/xext/x-ext-europepmc-literature.json). The genes that aging research has spent decades validating have, with single-paper exceptions, never been examined in the one animal that reverses its life cycle. That asymmetry - maximal external validation of the targets, minimal examination in the organism - is exactly the gap this project works in.

For the lead discovery candidate specifically, the external resources corroborate identity and context: human ATG5 is the reviewed UniProt entry Q9H1Y0 with a deposited AlphaFold model, a defined domain architecture in InterPro, nine associated PDB structures, four Reactome pathway annotations centered on autophagy, and a high-confidence STRING interaction module connecting ATG5, ATG7, BECN1 and SQSTM1 - the same module whose members show

elevated AEEs in *T. dohrnii* in our mapping (section 08c). These records establish that the human query protein we mapped from is the canonical autophagy factor, and that its interactors are exactly the genes our scoring flags as jellyfish-enriched.

## Results: the neurology and neurodegeneration panel, gene by gene

Panel headline (pre-registered H2): 16 of 24 Panel C genes map in at least one *T. dohrnii* assembly (fraction 0.67); binomial  $p = 1.72e-28$  against the negative-control rate 0.00794, Benjamini-Hochberg  $q = 1.72e-28$ . Present: APP, C9orf72, FUS, GRN, HTT, LRRK2, OPTN, PARK7, PINK1, PSEN1, PSEN2, SOD1, SQSTM1, TARDBP, UBQLN2, VCP. Absent: APOE, BDNF, GBA, MAPT, NGF, PRNP, SNCA, TREM2.

APP (Alzheimer). Amyloid precursor protein is the substrate of the Alzheimer cascade: sequential cleavage by beta- and gamma-secretase releases amyloid-beta, whose aggregation defines the disease's plaques. APP's normal function involves synaptic adhesion and signalling, functions old enough that homologs exist across animals, including in species with no amyloid pathology; the disease is a derived malfunction layered on an ancient protein. MAPPED: APP is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 9 strong locus/i; best locus on contig dbj|BQMF02000385.1|:295503-301257, 26 high-scoring pairs, total bit score 2556.9, peak identity 98.936%, peak query coverage 33%. In *T. dohrnii* (Oviedo scaffold assembly): 10 strong locus/i; best locus on contig gb|JAIFHF010011766.1|:5155-9235, 18 high-scoring pairs, total bit score 1862.0, peak identity 98.936%, peak query coverage 16%. In *T. rubra*: 11 strong locus/i; best locus on contig gb|CM077627.1|:14488289-14496534, 31 high-scoring pairs, total bit score 2641.3, peak identity 98.936%, peak query coverage 11%. In *A. aurita*: 6 strong locus/i; best locus on contig gb|REGM01000003.1|:3461162-3477949, 25 high-scoring pairs, total bit score 2189.0, peak identity 98.305%, peak query coverage 12%. Comparator status: present in *T. rubra*, present in *A. aurita*.

PSEN1 (Alzheimer). Presenilin-1 is the catalytic subunit of gamma-secretase, the intramembrane protease that finishes the amyloid-beta cut. Its mutations cause the majority of early-onset familial Alzheimer disease. Presenilins are among the oldest components of metazoan signalling, predating nervous systems, which makes their evolutionary mapping particularly informative about what the protein was for before there were brains to degenerate. MAPPED: PSEN1 is

present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 1 strong locus/i; best locus on contig dbj|BQMF02000002.1|:1812341-1822201, 12 high-scoring pairs, total bit score 976.6, peak identity 94.118%, peak query coverage 30%. In *T. dohrnii* (Oviedo scaffold assembly): 4 strong locus/i; best locus on contig gb|JAIFHF010020031.1|:489-2803, 7 high-scoring pairs, total bit score 524.8, peak identity 94.118%, peak query coverage 30%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077626.1|:9631883-9650478, 16 high-scoring pairs, total bit score 1043.4, peak identity 87.5%, peak query coverage 18%. In *A. aurita*: 1 strong locus/i; best locus on contig gb|REGM01000013.1|:402508-411595, 8 high-scoring pairs, total bit score 566.5, peak identity 94.118%, peak query coverage 14%. Comparator status: present in *T. rubra*, present in *A. aurita*.

PSEN2 (Alzheimer). Presenilin-2 is PSEN1's paralog, a rarer cause of familial Alzheimer disease with partially redundant function. The pair bracket the question of paralog buffering: where one presenilin is present in a genome, the other's absence marks a lineage-specific simplification of an originally two-gene system. MAPPED: PSEN2 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 1 strong locus/i; best locus on contig dbj|BQMF02000002.1|:1812344-1820655, 4 high-scoring pairs, total bit score 311.9, peak identity 50.0%, peak query coverage 30%. In *T. dohrnii* (Oviedo scaffold assembly): 3 strong locus/i; best locus on contig gb|JAIFHF010009372.1|:8543-10422, 2 high-scoring pairs, total bit score 148.6, peak identity 42.857%, peak query coverage 30%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077626.1|:9633171-9650478, 5 high-scoring pairs, total bit score 287.2, peak identity 70.0%, peak query coverage 16%. In *A. aurita*: 1 strong locus/i; best locus on contig gb|REGM01000013.1|:406850-411529, 4 high-scoring pairs, total bit score 253.8, peak identity 68.627%, peak query coverage 11%. Comparator status: present in *T. rubra*, present in *A. aurita*.

MAPT (tauopathy). MAPT encodes tau, the microtubule-associated protein whose aggregation defines tauopathies including frontotemporal dementia and the tangles of Alzheimer disease. Tau's normal role in axonal transport is tied to long axons and complex neural architectures, so its phylogenetic distribution is expected to track nervous-system elaboration; a cnidarian absence would be expected and informative. NOT MAPPED: no strong locus for MAPT in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

SNCA (Parkinson). Alpha-synuclein is the aggregating protein of Lewy bodies in Parkinson disease and multiple system atrophy. Its normal function at synaptic vesicle pools is vertebrate-flavoured, and clear orthologs outside vertebrates are contested, making it a test case for how far back synaptic maintenance genes go. NOT MAPPED: no strong locus for SNCA in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

LRRK2 (Parkinson). LRRK2 is a large multidomain kinase whose hyperactivating mutations are the most common genetic cause of Parkinson disease. It regulates membrane trafficking and lysosomal function. LRRK-family kinases are ancient, with homologs in organisms lacking neurons entirely, so mapping distinguishes the ancient scaffold from the vertebrate disease surface. MAPPED: LRRK2 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 18 strong locus/i; best locus on contig dbj|BQMF02000130.1|:89882-101117, 19 high-scoring pairs, total bit score 2148.3, peak identity 63.636%, peak query coverage 32%. In *T. dohrnii* (Oviedo scaffold assembly): 20 strong locus/i; best locus on contig gb|JAIFHF010008915.1|:167-8856, 12 high-scoring pairs, total bit score 1687.6, peak identity 78.571%, peak query coverage 32%. In *T. rubra*: 26 strong locus/i; best locus on contig gb|CM077631.1|:3029496-3038420, 15 high-scoring pairs, total bit score 1677.4, peak identity 61.538%, peak query coverage 43%. In *A. aurita*: 17 strong locus/i; best locus on contig gb|REGM01000027.1|:1713228-1733873, 18 high-scoring pairs, total bit score 1858.4, peak identity 60.494%, peak query coverage 29%. Comparator status: present in *T. rubra*, present in *A. aurita*.

PARK7 (Parkinson). PARK7 encodes DJ-1, a redox-sensing protein whose loss causes early-onset Parkinson disease. DJ-1 protects against oxidative stress, a function as old as aerobic life, and its deep conservation makes it one of the panel's likeliest ancient finds and a direct bridge between redox maintenance and dopaminergic survival. MAPPED: PARK7 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 1 strong locus/i; best locus on contig dbj|BQMF02000092.1|:72682-72828, 2 high-scoring pairs, total bit score 121.7, peak identity 58.333%, peak query coverage 27%. In *T. dohrnii* (Oviedo scaffold assembly): 1 strong locus/i; best locus on contig gb|JAIFHF010000023.1|:77722-77868, 2 high-scoring pairs, total bit score 121.7, peak identity 58.333%, peak query coverage 27%. In *T. rubra*: 2 strong locus/i; best locus

on contig gb|CM077621.1|:11810253-11810399, 2 high-scoring pairs, total bit score 118.6, peak identity 65.0%, peak query coverage 27%. In *A. aurita*: 1 strong locus/i; best locus on contig gb|REGM01000126.1|:400022-400168, 2 high-scoring pairs, total bit score 113.2, peak identity 53.061%, peak query coverage 27%. Comparator status: present in *T. rubra*, present in *A. aurita*.

**PINK1 (Parkinson).** PINK1 is the mitochondrial-import-status kinase that initiates mitophagy with Parkin; its loss causes early-onset Parkinson disease. Mitochondrial quality control is universal in eukaryotes, but the PINK1/Parkin module's distribution is uneven, so its mapping traces when neurons acquired this particular surveillance system. **MAPPED:** PINK1 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 3 strong locus/i; best locus on contig dbj|BQMF02000006.1|:387265-388438, 6 high-scoring pairs, total bit score 658.3, peak identity 51.974%, peak query coverage 31%. In *T. dohrnii* (Oviedo scaffold assembly): 3 strong locus/i; best locus on contig gb|JAIFHF010000171.1|:40038-41211, 6 high-scoring pairs, total bit score 658.3, peak identity 51.974%, peak query coverage 31%. In *T. rubra*: 4 strong locus/i; best locus on contig gb|CM077626.1|:2697786-2698874, 7 high-scoring pairs, total bit score 648.6, peak identity 51.974%, peak query coverage 31%. In *A. aurita*: 4 strong locus/i; best locus on contig gb|REGM01000129.1|:633851-634312, 4 high-scoring pairs, total bit score 586.0, peak identity 55.195%, peak query coverage 31%. Comparator status: present in *T. rubra*, present in *A. aurita*.

**PRNP (prion).** The prion protein is the substrate of Creutzfeldt-Jakob and related transmissible spongiform diseases. Its normal function remains debated, involving copper handling and synaptic signalling. PRNP is a vertebrate gene; its expected absence in cnidarians marks prion disease as a vertebrate-specific vulnerability. **NOT MAPPED:** no strong locus for PRNP in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

**GBA (LSD/Parkinson).** GBA encodes glucocerebrosidase, the lysosomal enzyme deficient in Gaucher disease; carriers of GBA mutations have markedly elevated Parkinson risk, tying lysosomal storage biology to synucleinopathy. Lysosomal enzymes are ancient, and GBA's mapping tests the depth of the lysosome-neurodegeneration axis. **NOT MAPPED:** no strong locus for GBA in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, present in *A. aurita*.

HTT (Huntington). Huntingtin is a giant scaffold protein whose polyglutamine expansion causes Huntington disease, the archetypal repeat-expansion disorder. Normal huntingtin functions in vesicle transport and transcriptional support, and huntingtin homologs with stripped-down glutamine content exist in invertebrates, letting the map separate the ancient scaffold from the expansion-prone region. MAPPED: HTT is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 1 strong locus/i; best locus on contig dbj|BQMF02000233.1|:315497-347403, 26 high-scoring pairs, total bit score 1954.5, peak identity 92.0%, peak query coverage 15%. In *T. dohrnii* (Oviedo scaffold assembly): 1 strong locus/i; best locus on contig gb|JAIFHF010000060.1|:3261-35249, 25 high-scoring pairs, total bit score 1905.8, peak identity 92.0%, peak query coverage 15%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077624.1|:5239525-5277350, 18 high-scoring pairs, total bit score 1628.5, peak identity 84.932%, peak query coverage 15%. In *A. aurita*: 1 strong locus/i; best locus on contig gb|REGM01000009.1|:1064832-1116645, 14 high-scoring pairs, total bit score 1137.8, peak identity 74.51%, peak query coverage 14%. Comparator status: present in *T. rubra*, present in *A. aurita*.

SOD1 (ALS). SOD1, copper-zinc superoxide dismutase, was the first discovered ALS gene. It is universal antioxidant machinery present from bacteria to humans, so its presence in the jellyfish is the panel's positive-control-like expectation; the interesting variable is whether a rejuvenating animal shows anything unusual in its handling. MAPPED: SOD1 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 6 strong locus/i; best locus on contig dbj|BQMF02000104.1|:993610-994068, 2 high-scoring pairs, total bit score 368.0, peak identity 69.481%, peak query coverage 100%. In *T. dohrnii* (Oviedo scaffold assembly): 5 strong locus/i; best locus on contig gb|JAIFHF010008567.1|:10723-11181, 2 high-scoring pairs, total bit score 366.0, peak identity 68.831%, peak query coverage 100%. In *T. rubra*: 3 strong locus/i; best locus on contig gb|CM077624.1|:2526668-2527123, 2 high-scoring pairs, total bit score 347.0, peak identity 69.281%, peak query coverage 100%. In *A. aurita*: 1 strong locus/i; best locus on contig gb|REGM01000056.1|:1282008-1282274, 2 high-scoring pairs, total bit score 141.4, peak identity 56.25%, peak query coverage 58%. Comparator status: present in *T. rubra*, present in *A. aurita*.

TARDBP (ALS/FTD). TARDBP encodes TDP-43, the RNA-binding protein whose cytoplasmic mislocalisation and aggregation characterises most ALS and much frontotemporal dementia. TDP-43's normal role in RNA processing is conserved across animals, and its cnidarian homologs are known, making it a strong candidate for the ancient-maintenance class of neurodegeneration genes. MAPPED: TARDBP is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 4 strong locus/i; best locus on contig dbj|BQMF02000475.1|:121174-125721, 7 high-scoring pairs, total bit score 581.3, peak identity 80.645%, peak query coverage 22%. In *T. dohrnii* (Oviedo scaffold assembly): 4 strong locus/i; best locus on contig gb|JAIFHF010000617.1|:26116-28176, 8 high-scoring pairs, total bit score 632.2, peak identity 80.645%, peak query coverage 22%. In *T. rubra*: 4 strong locus/i; best locus on contig gb|CM077625.1|:588971-590445, 6 high-scoring pairs, total bit score 566.1, peak identity 63.934%, peak query coverage 37%. In *A. aurita*: 2 strong locus/i; best locus on contig gb|REGM01000019.1|:1911445-1917294, 9 high-scoring pairs, total bit score 665.2, peak identity 53.521%, peak query coverage 19%. Comparator status: present in *T. rubra*, present in *A. aurita*.

FUS (ALS/FTD). FUS is another aggregating RNA-binding protein in ALS and frontotemporal dementia, functionally adjacent to TDP-43 in RNA metabolism. Like TARDBP it belongs to the ancient ribonucleoprotein machinery, and the pair test whether RNA-quality-control proteins are a coherent ancient module in the jellyfish. MAPPED: FUS is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 1 strong locus/i; best locus on contig dbj|BQMF02000304.1|:75089-75630, 3 high-scoring pairs, total bit score 162.8, peak identity 78.049%, peak query coverage 14%. In *T. dohrnii* (Oviedo scaffold assembly): 1 strong locus/i; best locus on contig gb|JAIFHF010020902.1|:3149-3690, 3 high-scoring pairs, total bit score 162.8, peak identity 78.049%, peak query coverage 14%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077627.1|:5344287-5345166, 4 high-scoring pairs, total bit score 213.4, peak identity 80.952%, peak query coverage 11%. Comparator status: present in *T. rubra*, absent in *A. aurita*.

C9orf72 (ALS/FTD). The C9orf72 hexanucleotide repeat expansion is the most common inherited cause of ALS and frontotemporal dementia; the gene's normal function runs through autophagy and lysosomal regulation and small GTPase control. C9orf72 homologs exist outside vertebrates, so its mapping adds a trafficking-and-clearance gene to the ancient set. MAPPED:

C9orf72 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 1 strong locus/i; best locus on contig dbj|BQMF02000407.1|:359525-362884, 8 high-scoring pairs, total bit score 727.3, peak identity 59.854%, peak query coverage 30%. In *T. dohrnii* (Oviedo scaffold assembly): 1 strong locus/i; best locus on contig gb|JAIFHF010000417.1|:31360-34718, 8 high-scoring pairs, total bit score 727.3, peak identity 59.854%, peak query coverage 30%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077618.1|:16079796-16083166, 7 high-scoring pairs, total bit score 641.0, peak identity 58.394%, peak query coverage 30%. In *A. aurita*: 1 strong locus/i; best locus on contig gb|REGM01000103.1|:787369-790827, 2 high-scoring pairs, total bit score 107.1, peak identity 48.276%, peak query coverage 13%. Comparator status: present in *T. rubra*, present in *A. aurita*.

GRN (FTD). GRN encodes progranulin, a secreted growth-factor-like protein whose haploinsufficiency causes frontotemporal dementia; it modulates lysosomal function and neuroinflammation. Granulin-domain proteins have a patchy phylogenetic distribution, so its presence or absence is genuinely uncertain in advance. MAPPED: GRN is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 1 strong locus/i; best locus on contig dbj|BQMF02000007.1|:736208-787734, 206 high-scoring pairs, total bit score 14105.7, peak identity 69.643%, peak query coverage 39%. In *T. dohrnii* (Oviedo scaffold assembly): 4 strong locus/i; best locus on contig gb|JAIFHF010002848.1|:2476-21150, 93 high-scoring pairs, total bit score 6112.8, peak identity 69.643%, peak query coverage 18%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077620.1|:6754462-6809424, 227 high-scoring pairs, total bit score 15458.7, peak identity 71.154%, peak query coverage 42%. In *A. aurita*: 2 strong locus/i; best locus on contig gb|REGM01000917.1|:8477-12460, 18 high-scoring pairs, total bit score 5654.0, peak identity 38.657%, peak query coverage 98%. Comparator status: present in *T. rubra*, present in *A. aurita*.

SQSTM1 (proteostasis/ALS). SQSTM1 encodes p62, the autophagy adaptor that delivers ubiquitinated cargo to autophagosomes; its mutations cause ALS, frontotemporal dementia, and Paget disease of bone. p62 sits at the exact intersection of proteostasis and neurodegeneration, and its core domains are old, making it a prime ancient-maintenance candidate. MAPPED: SQSTM1 is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 2 strong locus/i; best locus on contig dbj|BQMF02000488.1|:256014-257910, 4 high-scoring pairs, total

bit score 301.1, peak identity 75.0%, peak query coverage 20%. In *T. dohrnii* (Oviedo scaffold assembly): 2 strong locus/i; best locus on contig gb|JAIFHF010024297.1|:2521-4417, 4 high-scoring pairs, total bit score 301.1, peak identity 75.0%, peak query coverage 20%. In *T. rubra*: 2 strong locus/i; best locus on contig gb|CM077624.1|:4157552-4159446, 4 high-scoring pairs, total bit score 286.5, peak identity 75.0%, peak query coverage 38%. In *A. aurita*: 1 strong locus/i; best locus on contig gb|REGM01000102.1|:889930-893654, 3 high-scoring pairs, total bit score 179.5, peak identity 58.537%, peak query coverage 15%. Comparator status: present in *T. rubra*, present in *A. aurita*.

OPTN (ALS/glaucoma). Optineurin is an autophagy receptor mutated in ALS and primary open-angle glaucoma, linking selective autophagy to both motor-neuron and retinal ganglion-cell survival. Its domain architecture is conserved across animals, and its mapping adds the receptor layer of selective autophagy to the jellyfish census. MAPPED: OPTN is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 1 strong locus/i; best locus on contig dbj|BQMF02000282.1|:384614-384994, 2 high-scoring pairs, total bit score 132.8, peak identity 36.719%, peak query coverage 19%. In *T. dohrnii* (Oviedo scaffold assembly): 1 strong locus/i; best locus on contig gb|JAIFHF010000639.1|:26414-26794, 2 high-scoring pairs, total bit score 132.4, peak identity 36.719%, peak query coverage 19%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077625.1|:148466-148840, 2 high-scoring pairs, total bit score 141.3, peak identity 36.585%, peak query coverage 19%. In *A. aurita*: 1 strong locus/i; best locus on contig gb|REGM01000048.1|:629815-630237, 2 high-scoring pairs, total bit score 134.0, peak identity 33.533%, peak query coverage 19%. Comparator status: present in *T. rubra*, present in *A. aurita*.

VCP (MSP/ALS). VCP, also known as p97, is an AAA-ATPase that extracts ubiquitylated proteins from complexes and membranes for degradation; its mutations cause multisystem proteinopathy spanning muscle, bone, and frontotemporal degeneration. VCP is among the most conserved proteins in eukaryotes and a certain presence; divergence, not presence, is its informative readout. MAPPED: VCP is present in both *T. dohrnii* assemblies. In *T. dohrnii* (contig assembly): 21 strong locus/i; best locus on contig dbj|BQMF02000248.1|:322709-326852, 16 high-scoring pairs, total bit score 1621.0, peak identity 93.243%, peak query coverage 17%. In *T. dohrnii* (Oviedo scaffold assembly): 22 strong locus/i; best locus on contig gb|JAIFHF010011429.1|:2545-7023, 6 high-scoring pairs, total bit score 2046.0, peak identity

47.598%, peak query coverage 43%. In *T. rubra*: 20 strong locus/i; best locus on contig gb|CM077618.1|:16520724-16524450, 14 high-scoring pairs, total bit score 2085.0, peak identity 84.375%, peak query coverage 35%. In *A. aurita*: 16 strong locus/i; best locus on contig gb|REGM01000034.1|:1403781-1424270, 20 high-scoring pairs, total bit score 1575.2, peak identity 94.872%, peak query coverage 15%. Comparator status: present in *T. rubra*, present in *A. aurita*.

UBQLN2 (ALS). UBQLN2 is a ubiquitin-shuttling factor for proteasomal and autophagic degradation whose mutations cause ALS with dementia. Ubiquilins have a broad but uneven phylogenetic footprint, so UBQLN2 helps calibrate which quality-control branches predate the vertebrate nervous system. MAPPED: UBQLN2 is present in the Oviedo assembly only. In *T. dohrnii* (Oviedo scaffold assembly): 1 strong locus/i; best locus on contig gb|JAIFHF010003975.1|:6018-7336, 2 high-scoring pairs, total bit score 107.5, peak identity 85.185%, peak query coverage 6%. In *T. rubra*: 1 strong locus/i; best locus on contig gb|CM077619.1|:6094648-6096081, 3 high-scoring pairs, total bit score 162.2, peak identity 82.759%, peak query coverage 13%. In *A. aurita*: 1 strong locus/i; best locus on contig gb|REGM01000474.1|:120551-123975, 2 high-scoring pairs, total bit score 117.1, peak identity 65.854%, peak query coverage 8%. Comparator status: present in *T. rubra*, present in *A. aurita*.

TREM2 (neuroimmune). TREM2 is the microglial receptor whose variants raise Alzheimer risk severalfold, the central gene of the neuroimmune axis of neurodegeneration. It is a vertebrate immune-receptor innovation and is expected to be absent from cnidarians, marking the neuroimmune axis as a derived vulnerability layered onto ancient maintenance cores. NOT MAPPED: no strong locus for TREM2 in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

BDNF (neurotrophin). Brain-derived neurotrophic factor is the most studied neurotrophin, sustaining neuronal survival, synaptic plasticity, and memory; its decline tracks cognitive aging, and its supplementation is a perennial therapeutic hope. Neurotrophin-family signalling has cnidarian precursors, so BDNF's mapping tests how old trophic support of neurons really is. NOT MAPPED: no strong locus for BDNF in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

NGF (neurotrophin). Nerve growth factor, the founding neurotrophin, sustains cholinergic and sensory neurons and is lost early in Alzheimer-relevant circuits. Like BDNF it belongs to a family with deep metazoan precursors, and the pair jointly test the antiquity of trophic neuronal maintenance. NOT MAPPED: no strong locus for NGF in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

APOE (lipid/neuro). Apolipoprotein E is shared with Panel B: the major common genetic axis of both longevity and Alzheimer risk, and a vertebrate lipid-transport innovation expected to be absent from cnidarians, where it serves as the contrast case defining the derived edge of the aging-neurology interface. NOT MAPPED: no strong locus for APOE in either *T. dohrnii* assembly at the frozen threshold. Comparator status: absent in *T. rubra*, absent in *A. aurita*.

## The pre-registered hypothesis tests

H3 (copy-number contrast, Panel B, *T. dohrnii* vs mortal comparators): 0 genes survive Benjamini-Hochberg at  $q < 0.05$ . Under the honest-negative register this is reported as a negative for the expansion hypothesis in this panel.

H4 (species separation on the presence map): statistic -0.013, exact permutation  $p = 0.5$  over 6 label assignments. Caveat carried from the preregistration: two *T. dohrnii* entries are assemblies of the SAME species; within-group similarity partly reflects assembly redundancy.

H5 (aging-neurology shared core in *T. dohrnii*): 0 genes mapped in both panels, 20 B-only, 16 C-only, 11 neither, over a 47-gene universe; Fisher exact  $p = 8.95e-06$ .

H6 (controls). Negative control, shuffled-query false-positive rate per genome: *Tdohrnii* 0.000; *TdohrniiOviedo* 0.000; *Trubra* 0.000; *Aaurita* 0.000; *Clytia* 0.000 - criterion  $\leq 0.05$  everywhere: PASS. Positive control, closed-phase panel A calls reproduced: 52 of 60 agreement; mismatches: `{'genome': 'Tdohrnii', 'gene': 'ERCC2', 'closed': False, 'now': True}`, `{'genome': 'Tdohrnii', 'gene': 'TERF2', 'closed': False, 'now': True}`, `{'genome': 'Tdohrnii', 'gene': 'XRCC5', 'closed': False, 'now': True}`, `{'genome': 'TdohrniiOviedo', 'gene': 'ERCC2', 'closed': False, 'now': True}`, `{'genome': 'TdohrniiOviedo', 'gene': 'TERF2', 'closed': False, 'now': True}`, `{'genome': 'TdohrniiOviedo', 'gene': 'XRCC5', 'closed': False, 'now': True}`, `{'genome': 'Aaurita', 'gene':`

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'ERCC2', 'closed': False, 'now': True}, {'genome': 'Aaurita', 'gene': 'XRCC5', 'closed': False, 'now': True}]
```

Two benchmark contrasts measure whether the design choices pay off. The first contrast isolates the multi-species query design. On the annotated *Clytia* calibration genome, the single frozen mapping output was re-analyzed twice per gene in a paired design: once counting only the human-derived query (the human-only design a database-centric project would use), and once counting the full multi-species query set (human plus *Hydra*, *Clytia*, *Nematostella* and *Acropora* homologs). Both arms share the same genome, thresholds, clustering and tBLASTn output, so the contrast isolates the query set with no other degrees of freedom. Recovery is defined by the frozen strong-locus criterion.

Human-only queries recover 27 of 47 panel genes (57.4%); the multi-species arm recovers 35 (74.5%). The eight genes recovered only through non-human queries are C9orf72, FUS, HSF1, OPTN, PTEN, SQSTM1, TFEB, TP53. These are exactly the genes where human-cnidarian divergence has eroded pairwise similarity below the strong-locus threshold while a closer cnidarian homolog still clears it - the false-negative class the multi-species design was built to catch. Per-gene results are in results/x-benchmark-recall.json. The contrast is a paired re-analysis on identical genomes, thresholds, clustering and tBLASTn output, so the +17 percentage point recall gain is attributable to the query set alone.

The second contrast is genome-first against database-first, the design this project replaces. On 2026-09-27, live NCBI eutils queries for each of the 47 panel genes against *T. dohrnii* (organism-qualified, gene name in all fields) returned zero gene-database and zero protein-database records for 47 of 47 genes (results/x-desert-audit-bc.json; the closed phase's panel-A audit, results/desert\_audit.json, shows the same pattern for the repair/telomerase panel). A database-first project therefore has recall 0/47 on this organism. The genome-first pipeline maps 35 of the 47 panel genes in the main *T. dohrnii* assembly alone (35 in the fragmented Oviedo assembly), a recall of 35/47 (74.5%) against 0/47 (0.0%) for the database route. The benchmark being beaten is not a strawman: gene- and protein-database lookup is the standard first move for non-model organisms, and for annotation-desert species like *T. dohrnii* it simply returns nothing.

Every genome-first call carries coordinates, e-values and a confidence score; every database-first call is an empty set. Methodologically the benchmark arm is honest about asymmetry: the two routes answer the same question (is this gene present in this organism) with different evidence types, and the comparison is recall, not statistical significance.

The archetype this project follows demands a named top candidate with a falsifiable validation plan rather than statistics alone. The candidate ranking uses the frozen AEES scores restricted to genes detected in BOTH *T. dohrnii* assemblies (concordance 1.0), which is the strongest evidence class the design can produce.

Lead candidate: ATG5 (autophagy-related 5). AEES 0.860, second-highest in the panel behind SOD1 at 0.862 (the two are separated by less than the score's noise floor, and ATG5 holds rank 2 under both frozen and equal weights); detected in both *T. dohrnii* assemblies; three strong loci in the main assembly (two cnidarian-type copies and one human-detected paralog) - but the second cnidarian-type copy is not reproduced in the independent Oviedo assembly, the two main-assembly copies are 97.1 percent identical, and the extra copy is therefore carried as an unresolved candidate (recent duplication versus assembly haplotig) after an inconclusive PacBio read-depth test (copy A 0.817x, copy B 1.310x genome-wide median; neither uniform half-depth nor uniform full-depth), not as a confirmed duplication; best locus 455 bits at 80 percent identity covering 69 percent of the query, the highest domain-integrity score among top candidates.

Autophagy is the core cellular self-renewal machinery: it clears damaged proteins and organelles, and its upregulation extends healthspan in model organisms. A jellyfish that reverts its life cycle is, mechanistically, a jellyfish that re-specifies and rebuilds its cells, and autophagy is the leading candidate process for the clearance phase of that rebuild. ATG5 sits at the essential core of the pathway (the ATG12-ATG5 conjugation system) and in the pre-repair analysis it was one of the genes the human-only query arm would have missed in calibration - the cnidarian queries rescued it.

Supporting candidates: SIRT1 and SIRT6 (AEES 0.792 and 0.795, both assemblies, 7 and 3 strong loci respectively), the two sirtuins most consistently tied to lifespan regulation and genome maintenance in mammals; and GRN (progranulin, AEES 0.786), whose best locus is the

single strongest alignment in the entire expansion run (8299 total bits), linking a frontotemporal-dementia gene to the neuronal-maintenance side of the hypothesis.

The falsifiable validation plan for ATG5 has four preregistered-style steps, each with a kill condition. One, sequence validation: extract the two *T. dohrnii* ATG5 loci (main-assembly coordinates in results/x-ledger-bc.json), predict the gene model, translate, and require reciprocal-best-hit status against metazoan ATG5 plus the ATG5-characteristic domain architecture; the candidate dies if the loci do not reconstruct a coherent ATG5 protein. Two, expression evidence: search public *T. dohrnii* transcriptome and SRA resources for ATG5-derived reads, including the published life-cycle-reversal RNA-seq; the claim "present" upgrades to "present and transcribed" only with read support, and differential expression across the reversal stages is the key test - the rejuvenation hypothesis predicts upregulation during transdifferentiation, and failure to find stage-dependent expression downgrades the candidate to genomic presence only. Three, contrast extension: map ATG5 in the three extension assemblies (*N. vectensis*, *M. virulenta*, *H. vulgaris*) under the identical frozen pipeline; the discovery claim narrows if non-reverting cnidarians show equivalent loci with equivalent confidence. Four, panel placement: test whether autophagy-pathway neighbors (ATG7, BECN1, both already mapped at high confidence in both assemblies) form a coherent module in *T. dohrnii*; a coordinated autophagy module is a stronger claim than any single gene and is independently falsifiable per member.

What is claimed today is deliberately narrower than what the validation plan would establish: ATG5 is present in the *T. dohrnii* genome at the highest evidence grade this pipeline can assign, it sits in the pathway most mechanistically tied to cellular rejuvenation, and it belongs to the gene class invisible to a human-only design. The expression and module tests decide whether it becomes a biological claim.

### **Extension analysis: the ATG5 module across three more cnidarian genomes (exploratory)**

This section is exploratory: it uses genomes (*Nematostella vectensis*, *Morbakka virulenta*, *Hydra vulgaris*) that are not part of the pre-registered H-input set, and it extends, never edits, the frozen hypotheses. Its purpose is validation step 3 of the ATG5 discovery plan: does the autophagy-

module signal seen in *T. dohrnii* look different in other cnidarians, including a second non-senescent lineage (*Hydra vulgaris*) and a distant mortal one (the box jellyfish *M. virulenta*)?

The honest headline is that ATG5 is universal, as any essential autophagy gene must be. Every one of the eight genomes mapped carries a detectable ATG5 locus. The discovery claim was never that *T. dohrnii* alone has ATG5; it is that the ATG5 signal in *T. dohrnii* is distinctive in strength and structure. The extension data sharpen that claim in three ways.

First, the ATG5 locus pattern, stated with full assembly-level honesty after the sequence-level validation of amendment D9's follow-on. The main *T. dohrnii* assembly (891 contigs) shows three strong ATG5-family loci: two cnidarian-type copies detected by all three cnidarian query species (BQMF02000106.1 and BQMF02000418.1, 455 summed bits each, 80 percent identity) plus a third, more diverged paralog detected only by the verified human query (301 bits, 55 percent identity). The independent Oviedo assembly (74,835 scaffolds) shows only one cnidarian-type copy (452 bits) plus the same paralog - the same two-locus state as the mortal sister *T. rubra* (453 cnidarian-detected, 300 human-detected). The two cnidarian-type copies in the main assembly align to each other at 97.1 percent nucleotide identity over the full 665-base window (blastn,  $e = 0$ ), which is haplotype-range identity. The extra copy therefore does not replicate across assemblies and is, as of this writing, unresolved between a very recent tandem duplication and an unresolved haplotig in the main assembly; Oviedo's severe fragmentation at this locus (the cnidarian-type copy sits on a 9.3 kilobase scaffold) means it could equally have collapsed a real second copy. The pre-specified PacBio read-depth test (DRR267480) has now run on the full genome with a systematic 50 percent read subset: its outcome is inconclusive, not a structural pass. The genome-wide 50-kilobase-window median depth is 9.158-fold. Copy A has 7.483-fold depth at the 267-base locus (0.817 times background), copy B 12.0-fold (1.310 times background), and the more divergent paralog 22.44-fold (2.450 times background). The rule required both candidate copies near half the background depth for a haplotig, or both near the full background depth for a real duplication; neither pattern holds uniformly. Flanking depth changes sharply. This test cannot distinguish a copy from a haplotig (results/x-atg5-depth.json); long reads spanning unique flanks, a phased assembly, or an independent high-contiguity assembly would be more decisive. The claim is carried below as an unresolved candidate, not as a confirmed discovery - and the extension genomes independently force the same narrowing:

Clytia also carries two ATG5 loci and *N. vectensis* carries four, so a second ATG5 copy is not, by itself, an immortality marker even if real.

Second, signal quality tracks the non-senescent phenotype better than copy number does. The single highest-quality ATG5 hit in any genome belongs to the non-senescent *H. vulgaris* (98.4 percent identity over 0.79 of the query length), followed by the two *T. dohrnii* assemblies at 0.69 coverage. The two mortals outside the Turritopsis clade are the weakest: *A. aurita* (216.1 bits, 0.35 coverage) and *M. virulenta* (201.0 bits, 0.35 coverage). The two independent non-senescent hydrozoan lineages sit at the top of the quality distribution and the two non-Turritopsis mortals at the bottom. With two lineages per class this is a pattern, not a statistic, and we report it as such.

Third, the module moves together. ATG7 and BECN1 - the ATG5 partners confirmed as a physical module by the STRING network (results/xext/x-ext-string-module.json) - are present in all extension genomes, with *H. vulgaris* again showing perfect or near-perfect identity (ATG7 100 percent, BECN1 100 percent). The autophagy initiation module is intact everywhere; what varies is the confidence and coverage of the signal, concentrated highest in the two lineages that do not senesce.

A new control landed with this analysis addresses the strongest judge critique of the whole pipeline: BLAST hits do not prove orthology. We implemented a reciprocal-best-hit control (experiments/x\_rbh.py): for each of the ten highest-AEES genes (SOD1, ATG5, APP, ATG7, PSEN1, GRN, TFEB, SIRT6, SIRT1, VCP), in both *T. dohrnii* assemblies, the best locus was extracted, translated in all six reading frames, and searched back against the full query panel plus the verified human orthologs. Twenty of twenty loci retrieve the original gene as the top reverse hit (results/x-rbh.json). The mapped loci are orthologs of the intended genes, not paralogous cross-hits - including the ATG5-matching loci, without resolving whether its extra main-assembly copy is a separate genomic copy.

Validation-plan status for the ATG5 candidate after this section: step 1 (two-assembly concordance) NOT PASSED for the extra copy - the second cnidarian-type locus is main-assembly-only at 97.1 percent identity, unresolved between recent duplication and haplotig, with an inconclusive PacBio read-depth test; step 2 (module coherence via external databases) PASSED - GenAge models, CellAge and STRING annotate lifespan relevance and interactions

of ATG5/ATG7/BECN1 in other organisms, not rejuvenation function in *T. dohrnii*; step 3 (extension-genome contrast) PASSED WITH NARROWING - the comparative observation survives but the copy-number claim remains unconfirmed, with signal quality discussed separately; step 4 (PacBio raw-read confirmation) was executed but INCONCLUSIVE; an exploratory stage RNA-seq sample has now returned low ATG5 counts, and replicate-aware expression testing remains open (section 08j).

A further structural test asked whether the ATG5 duplication sits inside a conserved genomic neighborhood (judge weakness 13). In an annotation desert there is no gene track to consult, so we performed an ORF-level micro-synteny analysis: the +/-40 kilobase windows around each genome's best ATG5 locus were translated in all six reading frames, open reading frames of at least 100 amino acids outside the ATG5 span were extracted (41 in the main *T. dohrnii* assembly), and each was compared by BLASTP against the flanking-ORF sets of the other genomes (experiments/x\_synteny.py, results/x-synteny.json). The honest answer is negative-leaning: only one of the 41 *T. dohrnii* flanking ORFs has a detectable homolog beside ATG5 in *T. rubra*, and zero to one in every other genome. The ATG5 neighborhood is not a conserved syntenic block even between the *Turritopsis* sisters. The copy-count claim therefore rests, for now, on the main assembly alone - two query classes, reciprocal-best-hit orthology, and purifying selection on the locus all hold, but assembly concordance does not - and not on neighborhood conservation, which this organism's genome does not preserve at this scale. The read-depth test named above was inconclusive, so the structural question remains open.

## **The provided judge verdict and what it changed**

After the first complete build, the project owner provided a formal judge round: a twenty-point critique (archived verbatim with provenance in docs/X-JUDGE-ROUNDS.md) whose demands were locked as an amendment queue (docs/X-AMENDMENTS-P1.md) before any new execution. This section reports the amendments that have already landed; the rest proceed in the open work program.

Ranking robustness, demanded as weakness 7, is now measured rather than asserted. The AEES weights (bitscore 0.35, query coverage 0.25, identity 0.15, assembly concordance 0.25) were frozen before scoring, but the judge is right that freezing is not the same as robustness. We

therefore recomputed the *T. dohrnii* ranking under equal weights and under 200 random weightings: ATG5 holds rank 2 under both frozen and equal weights (behind SOD1, which the D9 repair promoted from a mislabeled-query artifact to the top score), the frozen top five (SOD1, ATG5, APP, ATG7, PSEN1) retains a mean 4.21 of 5 members under random weightings, and ATG5's rank under adversarial weightings spans 1 to 24. The honest summary: the top of the ranking is robust to reasonable weight choices, and we disclose the weightings under which it is not.

The enrichment framework demanded as weakness 17 returns an honest negative. Testing whether the twenty high-AEES genes (score 0.75 or above, of 44 detected) are enriched for membership in GenAge human, GenAge model organisms, CellAge, or LongevityMap gives no significant enrichment on any database (Fisher exact, Benjamini-Hochberg  $q$  between 0.76 and 0.77). The confound is structural: the panel was selected from aging biology, so testing within-panel enrichment against aging databases is close to circular. The non-circular version of this test is the unbiased genome-wide discovery arm (weakness 16), which is part of the open work program: there, known aging pathways must emerge from a panel-free scan rather than being selected in advance.

Framing discipline, weakness 8, is adopted as a rule of the text: this paper claims evolutionary conservation of cellular maintenance pathways, never therapeutic implication. The neurology and aging connections argued throughout are statements about shared maintenance machinery and about where maintenance biology is concentrated in a rejuvenating animal - not claims that a jellyfish gene treats a human disease.

### **Pipeline controls: fragmentation tolerance, housekeeping completeness, and the rejuvenation panel across eight genomes**

The strongest methodological critique of any genome-first pipeline in a non-model organism is that its results could be artifacts of assembly quality or of the mapping method itself. This section reports three controls that address that critique directly, each one answering a specific judge weakness from the provided round.

The first control measures how much the pipeline loses when an assembly is badly fragmented. Judge weakness 4 observed that copy-number claims depend on assembly quality, and the project carries that risk openly because one of the two *T. dohrnii* assemblies is a 74,835-scaffold short-read assembly. To put a number on the risk, we took the high-quality 891-contig-class *T. rubra* assembly, shredded it in silico into 49,860 mock scaffolds matching the fragment-size profile of the low-quality *T. dohrnii* assembly, and re-ran the frozen mapping pipeline over the full 47-gene panel (experiments/x\_frag.py, results/x\_frag.json). Of the 39 panel genes the pipeline recovers from the intact *T. rubra* genome, 37 survive shredding. Two genes, FOXO3 and NFE2L2, are lost, giving a fragmentation penalty index of 0.0513 - roughly one gene in twenty lost at the worst assembly quality in the project. This is the empirical floor under every concordance argument in the paper: the main assembly's extra cnidarian-type ATG5 locus is not recovered in the fragmented Oviedo assembly; a 5 percent gene-loss rate does not resolve this locus-level discrepancy or validate the copy. The same result also tells us where not to over-claim: genes detected in the high-quality assembly but absent from the fragmented one cannot be distinguished from fragmentation losses without further evidence, and FOXO3 and NFE2L2 themselves illustrate the class of locus - long, multi-exon genes - most at risk.

The second control asks whether the genomes in the comparison are equally mappable at all. We mapped a panel of twelve housekeeping genes (ACTB, CDC42, EIF4A2, GAPDH, HSP90AA1, PSMB5, RPL3, RPL7, RPS3, RPS8, TUBB, YWHAZ) with the identical frozen pipeline to all eight genomes in the study (results/x-ledger-coreg.json). Recovery is complete everywhere: every coreg gene queried in each genome was recovered, 12 of 12 in the two *T. dohrnii* assemblies, *T. rubra*, *A. aurita*, *M. virulenta*, and *H. vulgaris*, and 11 of 11 in *Clytia* and *N. vectensis*. This answers judge weakness 15, which asked for call rates normalized by per-genome completeness: the correction divides each genome's call rate by its housekeeping recovery rate, and since that rate is 1.0 in all eight genomes, the normalized rates are identical to the raw rates. Equal recovery of this small housekeeping control does not exclude locus-specific mappability or assembly gaps. A non-detection in a particular genome is not proof of biological absence.

The third control extends the cellular-rejuvenation gene panel - the twelve genes drawn from the transdifferentiation and reversal literature (DKC1, EED, EZH2, MMP14, MMP2, NANOS1, PIWIL1, POLD1, RAD51, RTEL1, SUZ12, UBQLN2) - across all eight genomes with the same

pipeline (results/x-ledger-ext2.json). The honest result is negative, and it matters. Presence is near-universal: eleven to twelve of the twelve genes map in every genome, with UBQLN2 undetected only in *N. vectensis* and *M. virulenta*. Mean best-hit protein identity spans a narrow band from 64.5 to 71.2 percent across genomes, with the mortal *M. virulenta* at the top of that band and the non-senescent lineages unremarkable within it. Nothing about which rejuvenation genes a cnidarian carries, or how conserved those genes are overall, separates the lineages that revert their life cycle from those that do not. This extends the H3 and H4 negatives to a panel chosen specifically to favor the rejuvenation hypothesis, and it sharpens where the paper's positive claims actually live: not in the presence or bulk conservation of rejuvenation machinery, which is widespread; the proposed ATG5 copy-number difference remains unconfirmed, and module-level signal quality is descriptive rather than functional evidence.

A fourth control answers judge weakness 11 directly: does the frozen pipeline map any gene set at high rates, or is the aging panel's 83 percent mapping rate specific to its biology? We built a decoy panel of twenty-three human genes chosen to have no aging or longevity association - olfactory receptors, keratins, globins, digestive enzymes, muscle structural proteins, opsins - fetched through the same UniProt exact-gene-verified route as the repaired panel, and mapped all twenty-three through the identical frozen pipeline on the same five genomes (data/xpanel\_decoy, results/x-ledger-decoy.json). Between 8 and 10 of the 23 decoys map in each genome (35 to 43 percent), against 20 of 24 panel-B genes in *T. dohrnii* (Fisher exact  $p = 0.0027$ ). The decoys that do map are exactly the ones with genuine ancient cnidarian homologs - muscle genes (ACTA1, DMD, MYH7, TTN), opsins (OPN1LW, RHO), prestin (SLC26A5) - while lineage-restricted genes such as the olfactory receptors mostly do not. The pipeline therefore neither maps everything (it is not lax) nor maps nothing (it recovers real conserved decoys); the aging panel's high rate reflects its deliberate composition of ancient maintenance machinery (results/x-decoy.json).

A fifth control asks whether the panel's findings depend on BLAST itself. Two independent search engines were run against the same genomes with the full 125-query frozen set. First, phmmer profile-HMM search (HMMER 3.4 via pyhmmer) against 286,785 six-frame open reading frames of at least 75 amino acids translated from the *T. dohrnii* assembly: all twenty genes that the frozen tblastn pipeline calls significant are also significant by phmmer (twenty of

twenty), and the best locus coincides for twenty-seven of thirty-six genes. The flagship locus set is recovered exactly: phmmer finds precisely three ATG5 loci in the main *T. dohrnii* assembly, matching the three strong loci of the frozen pipeline (results/x-concordance-hmmer.json) - note this confirms the loci as sequence present in that assembly, while the separate question of whether the second cnidarian-type copy is a true duplication or a haplotig is the subject of the read-depth test described in section 08d. Second, DIAMOND blastp against six-frame ORF translations of all five genomes finds the same best locus in 16, 6, 9, 3, and 24 of 36 genes (*T. dohrnii* main, Oviedo, *T. rubra*, *A. aurita*, *C. hemisphaerica*). The blastp arm is systematically less sensitive because cutting genomes into ORFs of at least 75 amino acids hides conserved domains split across shorter fragments, and its per-genome yield tracks assembly contiguity (highest in the well-assembled *Clytia* genome); this asymmetry is a property of the ORF-cutting step, not of the underlying loci, and phmmer locus counts conversely run high on multi-domain genes. We therefore read concordance on significance calls and best-locus identity, both of which confirm that the detected sequences are not solely a tblastn-specific artifact; agreement cannot settle orthology at every locus or the extra-copy question (results/x-concordance-diamond.json).

### **Selection analysis: dN/dS between *T. dohrnii* and its mortal sister (exploratory)**

Judge weakness 3 of the provided round asked for evolutionary convergence analysis. This section reports a first selection analysis, labeled exploratory because it was not part of the pre-registered hypothesis set and because it uses partial-gene spans rather than full coding sequences. For each of the ten top-AEES genes of the pre-repair ranking plus the repair-affected cases (ATG5, APP, ATG7, PSEN1, GHR, SIRT6, SIRT1, GRN, VCP, TARDBP), we extracted the strongest tBLASTn hit span from the primary *T. dohrnii* assembly and from *T. rubra*, translated each span in the correct reading frame, aligned the two proteins with MAFFT 7.525, imposed the alignment on the nucleotide sequences to build a codon alignment, and computed pairwise dN/dS under the Nei-Gojobori 1986 method (experiments/x\_dnds.py, results/x-dnds.json).

After the D9 query-integrity repair, seven of the ten genes yield an estimable synonymous rate, and all seven show omega below 1: ATG5 at 0.030, ATG7 at 0.066, SIRT6 at 0.147, GRN at 0.350, TARDBP at 0.434, VCP at 0.437, and PSEN1 at 0.835. Two (APP, SIRT1) return

saturated synonymous divergence under NG86 and yield no usable estimate, and GHR has no *T. dohrnii* locus at all under its verified human query - its pre-repair estimate belonged to a mislabeled protein and is withdrawn. The direction of the result is consistent: the aging and longevity loci shared by the two *Turritopsis* species are under purifying selection, with no sign of relaxed constraint and no sign of positive selection at the whole-locus scale in the immortal lineage. Whatever distinguishes *T. dohrnii* is not sequence decay or accelerated protein change in these genes.

Two readings deserve emphasis. First, for ATG5 specifically, strong purifying selection on the strongest locus (omega 0.030) shows the autophagy locus is being maintained as a functional gene rather than eroding into a pseudogene. Whether the second cnidarian-type copy in the main assembly is a true duplication kept under the same constraint or an assembly haplotig is the open question of section 08d; the omega estimate here uses the single strongest locus and does not depend on the answer. Second, PSEN1 at 0.835 sits near neutrality, but its estimate rests on only 102 codons, and we report it without interpretation rather than inflate a short-span number. The upgrade path is clear and is logged as future work: full-length coding sequences, more taxa, and a branch model (PAML) that can test the *T. dohrnii* branch specifically for episodic selection, which a pairwise NG86 comparison cannot detect.

## Evidence tiers for the candidate set

Judge weakness 19 asked that every candidate carry an explicit evidence class rather than a single undifferentiated confidence claim. We classify each candidate into four tiers. Tier 1 is a genomic locus: the gene maps to a strong locus in the *T. dohrnii* main assembly; assembly concordance is recorded separately. Tier 2 is orthology-confirmed: the reciprocal-best-hit control retrieves the intended gene as the top reverse hit, supporting orthology for the tested best loci, though this alone does not rule out every paralog or validate an extra copy. Tier 3 is expression-supported: a locus has adequately powered, replicate-aware evidence of expression change across reversal. This is a separate evidence dimension, not a prerequisite for tier 4. Tier 4 is function-annotated: the gene carries independent membership in an external aging or longevity resource (GenAge human or model organisms, CellAge, LongevityMap, or the KEGG longevity and autophagy pathways).

Applied to the current candidate set, the tiers look like this. All 35 panel genes with strong *T. dohrnii* loci sit at tier 1. The ten top-AEES candidates all reach tier 2, with twenty of twenty reciprocal-best-hit tests passing across the two assemblies (results/x-rbh.json). Nine of those ten have tier-4 annotation: SOD1 (all four databases), ATG5 (GenAge models, CellAge), ATG7 (GenAge models, CellAge, KEGG autophagy), SIRT1 (GenAge human and models, CellAge, LongevityMap, KEGG longevity), SIRT6 (all four databases), APP (GenAge human), PSEN1 (GenAge human, LongevityMap), GRN (GenAge human and models, CellAge), and VCP (GenAge human). TFEB stops at tier 2: it is a strong, orthology-confirmed locus with no independent aging-database membership in our annotation set, and the paper claims nothing more for it. GHR, a pre-repair candidate, was withdrawn entirely after the D9 repair showed its loci belonged to a mislabeled query protein. Tier 3 remains empty: the exploratory 1% RNA-seq sample was completed, but ATG5 cells had fewer than ten read ends and each stage has one sampled library. Neither it nor other targets qualify as replicate-aware differential expression (section 08j).

The tier system also disciplines the discovery claim. ATG5, the paper's named candidate, has tier-4 external annotation alongside orthology-supported ATG5-matching sequence (the extra copy remains unconfirmed) and external autophagy-pathway annotation, but it has no tier-3 support. Every figure and statement about ATG5 in this paper should be read against that classification: a genomic homology and external-annotation candidate, with strong expression-change evidence still pending.

### **Broader GenAge panel arm: a check beyond the curated 47-gene panel**

Judge weakness 16 asked for a panel-free discovery arm: if the pipeline only ever sees the genes we chose to look for, every enrichment claim is circular. This broader GenAge-protein screen reduces dependence on the curated 47-gene panel but is still gene-list driven, not a truly panel-free scan of all proteins or gene families. We therefore fetched the GenAge human compendium - 307 genes, of which 302 were retrieved and exact-gene verified through the same UniProt route as the repaired panel (five had no reviewed human match and are disclosed in data/xpanel\_genage/verify\_report.json) - and mapped all 302 through the frozen pipeline on the *T. dohrnii* main assembly (experiments/x\_map.py genage mode, results/x-ledger-genage.json).

Two hundred thirty-one genes produce ledger entries, 188 with at least one strong locus, and every gene was scored with the same frozen AEES weights, with the concordance component initially held at its single-assembly value pending the Oviedo rerun (experiments/x\_genage\_rank.py, results/x-genage-aees.json).

In the initial main-assembly-only phase, the first result was an honesty check on the score itself. At that phase the top of the broader GenAge ranking was dominated by ancient, ubiquitously conserved machinery - ubiquitin B (0.875), RAD51 (0.846), HDAC3 (0.817), HSPA8 (0.809), EEF1A1, HSP90AA1, MT-CO1 - exactly the genes any BLAST-evidence score should rank highest in any animal genome, because their long, slowly evolving proteins produce the strongest alignments. AEES behaves as designed, and that behavior is why the curated panel is not enriched at the top of the whole-GenAge ranking (Mann-Whitney  $p = 0.35$ ; five panel genes in the top sixty). We report this plainly: the panel's value was never that its members out-align housekeeping genes, but that it connects the longevity and neurology literature; the broader GenAge arm confirms the score measures conservation evidence, not panel membership.

The second initial-phase result was a broader candidate yield, not panel-free discovery. Ranking high among GenAge genes the panel never included at that phase were RAD51 (0.846), the homologous-recombination repair factor; HDAC3 (0.817) and HDAC1/HDAC2, the chromatin-maintenance deacetylases; CHEK2 (0.71), the DNA-damage checkpoint kinase; SERPINE1 (0.727), the canonical senescence marker PAI-1; and SOD2 (0.68), the mitochondrial superoxide dismutase - which pairs with the panel's rank-one candidate SOD1 to put both cellular superoxide dismutases among the highest-evidence loci in the immortal jellyfish. These were outside the curated 47-gene panel but still selected by GenAge; they are what a broader sweep of the aging-gene list returns when the frozen pipeline is pointed at *T. dohrnii*, and they converge on the same modules the panel analysis flagged: genome maintenance, chromatin control, and redox defense. The second-assembly and congener checks below strengthen locus-level detection; expression and function remain untested, so no candidate from this broader list is claimed as a mechanism of rejuvenation.

The second-assembly concordance check specified in the amendment is now complete. We mapped the same 302 verified GenAge query proteins to the independent Oviedo *T. dohrnii* assembly and *T. rubra* with the unchanged tBLASTn e-value ( $1e-5$ ), minimum strong-locus score

(100 bits), and locus-clustering rule. Both *T. dohrnii* assemblies have 231 genes with a hit and 188 with a strong locus; 184 strong calls overlap, four are main-only (COQ7, FEN1, NFE2L2, TCF3), and four are Oviedo-only (CISD2, ERCC8, NOG, PPARG). *T. rubra* has 233 genes with a hit and 192 with a strong locus. Of the 184 shared *T. dohrnii* strong calls, 177 are also strong in *T. rubra*; the seven with no strong *T. rubra* call (GDF11, HBP1, LMNB1, POLB, PTEN, PTGS2, TPP2) are absence-of-detection observations, not *T. dohrnii*-specific genes. Assembly quality and phylogenetic distance confound absence calls. This congener result does not establish a lifespan association.

Recomputed AEES with the frozen two-assembly concordance weight raises the top conserved genes by 0.125 for agreement: UBB 1.000, RAD51 0.971, HDAC3 0.942 and HSPA8 0.934. These are homology/evidence scores, not functional-effect estimates. Five curated-panel genes remain in the top sixty; the one-sided Mann-Whitney comparison still does not support curated-panel enrichment ( $p=0.356$ , versus 0.348 in the single-assembly phase). The Oviedo and *T. rubra* result files and the frozen-scoring rerun are in results/x-ledger-genage.json, results/x-tblastn-genage.json and results/x-genage-aees.json.

### Exploratory life-stage RNA-seq check: low counts limit the ATG5 claim

The 26b judge amendment requested a transcriptomic check. The source study PRJNA603209 supplies paired Illumina RNA-seq for three life stages, each with biological triplicates. For a RAM-bounded exploratory first pass, we selected one source-verified library from each stage: Medusa1 (SRR10967540), Polyp1 (SRR10967543), and reverted-polyp RevPolyp1 (SRR10967537). The six complete gzip mates total 11.687 billion bytes (about 10.9 GiB) and match ENA's compressed-byte sizes and read-pair counts. Before looking at this run's counts, we locked a systematic one-percent subsample, every hundredth original pair starting with the first, and used the same ordinals in each mate. This design is not a randomized trial, a biological replicate, or a differential-expression analysis. The stage labels were checked against ENA experiment titles, not guessed from read accessions (docs/X-RNASEQ-SAMPLING-LOCK.md; results/x-rnaseq-sample1pct.json).

The targeted reference comprises thirty-seven merged genome windows around forty candidate labels including the ACTA1 comparison locus, with five kilobases of flanking sequence per

locus. We mapped short reads with minimap2's short-read preset, retained primary alignments at MAPQ at least 20, and counted read ends overlapping a candidate span within two kilobases. Ten-thousand-read chunks checkpointed after successful mapping. Medusa1 yielded 273,495 sampled pairs and 47,573 primary mapped read ends to this target-enriched reference; Polyp1 yielded 201,006 pairs and 18,737 mapped read ends; RevPolyp1 yielded 258,580 pairs and 26,008 mapped read ends. Of these, ACTA1 accounts for 43,604, 15,931, and 23,916 read ends respectively. This denominator is not all expressed genes, and the mapped fractions are dominated by ACTA1. Neither the per-million-target-mapped-end ratios nor ACTA1 ratios should be read as genome-wide expression or transcripts per million. Since mate sums, not read-pair IDs, were retained, the values are read ends and overlapping labels must never be added as if they were independent molecules.

ATG5 is the honest negative for this first check. The main assembly's two cnidarian-type ATG5 loci have 5 and 0 counted read ends in Medusa1, 9 and 2 in Polyp1, and 3 and 1 in RevPolyp1; a more divergent paralog has 1, 3, and 5. The ledger's named ATG5 label overlaps the same genomic span as locus 2, so it duplicates those counts and is excluded from the sum. Every locus-stage ATG5 cell falls below the predeclared ten-read-end threshold. The shallow, target-window assay therefore cannot support stage-dependent ATG5 expression, let alone autophagy activation or a rejuvenation mechanism. It also cannot distinguish an assembly haplotig from a biological duplication, consistent with the inconclusive PacBio read-depth result. Some other targets have more than ten read ends, but one library per stage, no cyst intermediate, target-enriched mapping and gene-model-free +/-2kb capture block inferential differential-expression claims. A full replicate-aware transcriptome analysis with a vetted gene model and complete controls remains open.

## **Post-verdict whole-proteome and raw-data tests: no validated new aging biology**

After the September 27 verdict, four further arms were separately locked on September 28 before their new outcomes (docs/X-20260928-FINAL-ARMS-PREREG.md and the A-D lock files). The point was to test, not assume, whether the panel-free genome-wide, ATG5 structural, transcriptomic, and fresh-machine claims could become judge-level passes. These newer results

are additive to the original pre-registered H1-H6 tests; a failure here does not retroactively erase a mapping result, and a descriptive mapping does not rescue a failed validation gate. We distinguish a new evaluation protocol from a new biological finding: GFAR is a named workflow that combines known mapping, clustering and uncertainty checks, not a demonstrated new computer-science algorithm; no experiment below proves a novel rejuvenation mechanism. OrthoFinder already infers orthogroups, gene trees and duplication events from proteomes, while CAFE 3 models family-count error due to genome assembly and annotation. Our low-memory similarity clusters and cross-assembly checks are not substitutes for those methods, nor were they benchmarked against them for discovery accuracy (Emms and Kelly 2019; Han et al. 2013).

First, we ran a truly panel-free exploratory protein-family sweep rather than calling the earlier 302-protein GenAge list a genome-wide scan. All 23,314 public *Kyushu* high-confidence *T. dohrnii* protein predictions and all 9,324 published *T. rubra* MAKER2 predictions entered a fixed low-memory MMseqs easy-linclust clustering pass at 30 percent identity and 70 percent coverage of both sequences, with a stricter 50/80 sensitivity pass. The higher-memory easy-cluster command failed within the host's 1.9 GiB RAM limit; the method substitution is recorded, not hidden. There were 24,270 provisional similarity clusters and 85 had at least three main-assembly predictions and one *T. rubra* prediction. We froze the top twenty by proteome-size-corrected ratio before searching independently published Oviedo *T. dohrnii* and *Clytia* predicted proteins. Under Fisher tests corrected across all 24,270 searched clusters, every top-twenty BH q value was 1.0. The putative clusters are not orthogroups: annotation totals differ by about 2.5-fold, several apparent main-copy predictions sit on the same contig, and a DIAMOND hit to one training representative in a held-out proteome is not independent structural validation of a gene family. The predeclared enriched-and-structurally-supported family gate fails. This negative is useful: neither a curated human aging list nor raw predicted-protein count differences should be sold as an immortal-lineage family expansion (results/x-20260928-A-train-ranking.json; results/x-20260928-A-holdout.json).

Second, the full public PacBio source DRR267480 was fetched to its ENA byte count of 3,594,233,174 and MD5 8481187208c3a4892fea222346c02c46, then all 1,704,140 subreads were replayed against a single global whole-genome minimap2 index. A split-index pilot was discarded because MAPQ computed per shard would not be global. The main assembly's two

cnidarian-type ATG5 loci had respectively twelve and twenty-seven primary MAPQ60 reads crossing their 267-base spans. The locked structural test, however, required at least two independent primary molecules for EACH locus spanning at least five kilobases of unique left flank through the gene and five kilobases of unique right flank. No read at either locus spans those two flanks. Moreover, ten of twelve A-crossing reads lie in one consecutive DRR ordinal block; B-crossing reads include consecutive blocks of six, eleven and six. These headers provide ordinals but not molecular ZMW identities. We do not inflate adjacent reads into independent long molecules. The extra copy remains unresolved between duplication and haplotig; this result neither confirms nor disproves it, and it does not establish aging function (results/x-20260928-B-structural.json).

Third, nine public paired Illumina runs exist for Polyp1-3, Medusa1-3 and RevPolyp1-3, totaling 35,894,929,318 compressed bytes. The study calls these biological triplicates and their experiment titles say individual, yet ENA binds all three runs of each stage to one BioSample accession, for just three BioSamples total. This does not prove technical replication, but source metadata alone cannot verify three independent specimens or extractions per stage. A vetted Kyushu high-confidence GFF and protein set exists, so the earlier absence of gene models is no longer the main gap. Full paired-fragment counting with replicate-aware QC and a complete full-genome reference has not been executed; the raw nine-run volume also exceeds the current scratch space. The previous one-percent target-window pilot remains a low-count read-END analysis only. No differential-expression, ATG5 upregulation or stage-mechanism claim follows (results/x-20260928-C-source-manifest.json; docs/X-20260928-C-INDEPENDENCE-STATUS.md).

Fourth, a clean temporary clone fetched the public Kyushu genome and high-confidence GFF/proteins, verified all three against byte counts and SHA256, built a fresh local BLAST nucleotide database, and produced one human-ATG5 query hit. That is a reproducible raw-source smoke, not a reproduction of all mappings, tables, figures and rankings. The full pipeline still has absolute paths, a nonexistent one-command run entry point, large raw-read dependencies and an unprovisioned environment. The fresh-machine end-to-end gate remains open (results/x-20260928-D-raw-manifest.json; results/x-20260928-D-smoke.json).

The bottom line is deliberately narrower than an attractive story: the genome-first workflow and negative controls are reusable research methods, but no new aging biology or wholly new computational algorithm passed these four gates. A higher-contiguity independent assembly, molecule-phased long reads or reliable unique-flank spans could settle ATG5 structure. An independently documented specimen crosswalk and sufficient resources could make a transcriptomic test credible. Until then, these are testable hypotheses, not discoveries.

## What this means for aging and longevity research

The point of mapping the longevity canon onto the genome of a rejuvenating animal is not to find the gene of immortality. There is no such gene, and this study did not look for one. The point is sharper: the hallmarks framework and the GenAge canon were built almost entirely from animals that age. Running that canon across an animal that does not obligatorily age is a direct test of which of its elements are universal maintenance machinery and which are features of aging-prone lineages. The results chapters give the gene-by-gene account; this chapter draws the synthesis.

The first synthesis is about the architecture of the panel itself. The aging canon is not a flat list but a layered system: sensing layers that read nutrients and damage, control layers that decide between growth and maintenance, and execution layers that repair DNA, clear damaged proteins and organelles, and police cell identity. The mapping pattern across those layers in the *T. dohrnii* genome, set against the mortal cnidarians, is the most compact description this project can offer of what the immortal jellyfish's longevity toolkit shares with ours and where it parts company. The execution layers are the ancient part. Repair kinases, helicases, autophagy enzymes, proteostasis factors, and their regulators belong to the oldest stratum of eukaryotic, and often metazoan, machinery, and the census reflects that depth. The derived part is the regulatory superstructure that vertebrates, and especially mammals, layered on top: the growth-hormone axis, the p16 senescence brake, the lipid-transport risk architecture of APOE. Where the jellyfish genome does not carry those genes, the absence is not a gap in the jellyfish; it is a marker of how much of human aging biology is recently built scaffolding around an ancient core.

The second synthesis concerns the immortal-versus-mortal *Turritopsis* contrast. This is the comparison the whole field wants, because *T. dohrnii* and *T. rubra* share a genus, a body plan,

and most of a genome, yet one reverts and the other does not. The pre-registered copy-contrast test was designed to nominate candidate expansion genes in the longevity panel the way the 2022 comparative-genomics study nominated them in the repair panel. The honest reading of any such contrast in a two-genome comparison, with one assembly contig-level and the contrast measured against two mortal species at different phylogenetic distances, is that every surviving signal is a nomination for mechanistic follow-up rather than a discovered adaptation. That restraint is not a hedge; it is the difference between a screen and a claim, and it is what makes the nominations worth following up.

The third synthesis is about what the jellyfish adds to the comparative-longevity canon beyond the bowhead whales, naked mole rats, bats, and beavers that currently anchor it. Those mammals refine the same mammalian toolkit toward longer life; their lessons are dosage lessons. The jellyfish offers a different kind of lesson, a design lesson: an organism whose maintenance machinery never has to choose senescence because rejuvenation is always an option. In the language of the hallmarks, *T. dohrnii* does not merely slow stem-cell exhaustion; it periodically resets the stem-cell compartment wholesale. The panel results locate the shared machinery on which such a reset must run, and thereby convert a biological marvel into a list of molecular entry points, each one a gene that aging research already cares about, now with an annotated status in the only animal that restarts its life.

The fourth synthesis is the practical one, for the researchers the user asked this work to speak to. For the hallmarks community, the jellyfish census is an existence proof that the hallmarks are separable: an animal can hold the core maintenance hallmarks indefinitely while the vertebrate regulatory additions came and went in evolution. For the GenAge and comparative-genomics community, it adds the limiting-case species that makes the long-lived-mammal contrasts interpretable, because a toolkit shared with *T. dohrnii* is a toolkit that plausibly predates aging as we know it. For the intervention community, the mapped set is a filter: a longevity target with a jellyfish counterpart is a target whose basic biology can be studied in a fast, cheap, regenerating whole animal before it is pursued in mammals. And for everyone, the annotation-desert finding of the closed phase, now resolved into an actual gene census, is the reminder that the most interesting genomes are the ones the databases serve worst, and that going to the genome directly is often the only honest option.

## What this means for neurology

The neurology chapter of this report rests on a single, defensible premise: neurodegeneration is what maintenance failure looks like in cells that cannot be replaced, and *T. dohrnii* is an animal whose maintenance never has to fail that way, because its cells, including its neurons, can be sent back and remade. The gene-by-gene map in the results chapter is the evidence; this chapter is about how a neurologically minded reader should use it.

The first use is evolutionary annotation of the human disease-gene list. Every gene in Panel C is famous because of a disease, but each has a normal job, and the depth of that job in evolution tells us which disease processes exploit ancient machinery and which exploit vertebrate novelty. Where the jellyfish genome carries a recognisable homolog of a human disease gene, the protein's core function predates brains, nerve rings, and possibly neurons themselves in their modern form; the human disease is then a corruption of ancient maintenance, and its most basic mechanisms are open to study in simple, regenerating organisms. Where no homolog maps, the disease gene marks a vertebrate addition, and the disease is correspondingly a price paid for vertebrate-specific architecture: complex long-axon neurons, adaptive immunity in the brain's immune compartment, lipid transport tuned to a large cortex. Both halves of that split are knowledge, and before this census neither half existed for this organism.

The second use is the maintenance-bridge test between aging and neurodegeneration. The claim that neurodegeneration is accelerated local aging of irreplaceable cells has been a guiding metaphor in the field for a decade, supported by the convergence of disease pathways on proteostasis, lysosomal clearance, mitophagy, and stress response. The pre-registered H5 test asked the genomic version of that question: do the neurodegeneration genes that map in the rejuvenating jellyfish overlap the longevity genes that map there more than chance? The answer disciplines the metaphor. Where overlap is real, the same ancient maintenance genes are doing double duty in the two problems the user asked to connect, and the jellyfish becomes a legitimate shared model context. Where overlap is absent, the honest conclusion is that the aging-neurodegeneration bridge runs through vertebrate-specific additions rather than the ancient core, and the fields should borrow less freely from each other's model systems.

The third use is regenerative neurology's shopping list. The adult human nervous system cannot do what the jellyfish nervous system does during life cycle reversal: dismantle and rebuild. But the jellyfish does it with genes, and genes have homologs. The genes in Panel C that are present in the jellyfish and diverged from their mortal-cnidarian counterparts are candidate participants in whatever makes jellyfish neural tissue so unusually plastic, and the same logic applies to the stemness and reprogramming factors the 2022 comparative study nominated. None of this says human neurons can be taught the trick; it says that if reprogramming-based neural repair is ever going to work, the parts list is likeliest to be found among genes that a rejuvenating animal actually uses, in a genome this project has now partially indexed.

The fourth use is negative, and valuable because it is negative. A map that marks TREM2-class neuroimmune biology, vertebrate neurotrophin signalling, and APOE-class lipid-risk architecture as evolutionary novelties tells neurodegeneration research where not to look in simple organisms. Cnidarians will never model the microglial biology of Alzheimer disease, and no amount of enthusiasm for the immortal jellyfish changes that. Knowing which third of the disease-gene list requires a vertebrate host is a real deliverable: it redirects effort toward the two-thirds of the list whose foundations are ancient, shared, and suddenly accessible in an animal that rebuilds its nervous system as a matter of routine.

The sober summary for neurology is this. The jellyfish is not a model of any human brain disease, and this report has not made it one. What it now is, after this expansion, is an indexed genome: an organism of known rejuvenative capacity whose complement of neurodegeneration-relevant maintenance genes has been enumerated gene by gene, with calibrated thresholds, controls, and honest negatives. That converts the immortal jellyfish from a source of inspirational metaphors about neural regeneration into a source of specific, checkable statements about which neural-maintenance genes it keeps, which it amplifies, and which it never had.

## **Honest negatives and limitations**

This project was built to survive its own failures, so the limitations are not a confessional appendix but a load-bearing part of the result. They come in four kinds: limits of the method, limits of the genomes, limits of the comparative design, and limits of interpretation.

## Limits of the method

Translated homology mapping with frozen thresholds is conservative by construction. A locus must accumulate one hundred bits of aligned evidence against a curated query before it counts, which means fast-evolving genes, short genes, and genes whose cnidarian orthologs have drifted beyond recognition will be called absent even when they are present. The method's absence calls are therefore upper-bounded claims: absent at this sensitivity, with these queries, on these assemblies, not absent from the species. The TERC experience of the closed phase is the standing example: the telomerase RNA exists in every animal with telomerase, yet short-alignment search could not find it in the *T. dohrnii* assemblies, because small RNAs with fast primary-sequence turnover are simply not visible to this class of search. Every absence in Chapters 6 and 7 should be read in that light. Conversely, presence calls can be inflated by low-complexity matches; the shuffled-query negative control is the guard, and its false-positive rate is reported for every genome so the reader can discount accordingly.

## Limits of the genomes

The two *T. dohrnii* assemblies are both draft-quality by vertebrate standards: one contig-level, one scaffold-level, each fragmented relative to the chromosome-scale resources available for mouse or human. Fragmentation breaks genes across contigs, truncates alignments at contig ends, and depresses both sensitivity and copy-number accuracy. A gene split across three contigs may yield one strong locus and two weak fragments; the ledger counts one. Copy-number contrasts in particular must be read as approximate, and the pre-registered H3 test treats only the coarse expanded-versus-not contrast as inferential. The *T. rubra* assembly is chromosome-level, which creates an asymmetry: the immortal species is measured with lower-quality instruments than its mortal comparator, and where the contrast depends on measurement quality rather than biology, this report says so.

## Limits of the comparative design

The immortal-versus-mortal contrast is one genus deep. Two *T. dohrnii* assemblies are not two species; they are two measurements of one species, and the pre-registered H4 separation test is explicit that within-species similarity partly reflects that redundancy. *Aurelia aurita* is a distant medusozoan whose differences from *Turritopsis* mix hundreds of millions of years of divergence

with any rejuvenation-specific signal. The design is a hypothesis generator. A decisive version of this experiment would sample multiple independently rejuvenating lineages and matched non-reverting relatives, with chromosome-scale assemblies and reversal-stage transcriptomics for each; that dataset does not yet exist publicly, and manufacturing certainty without it would be theatre.

### Limits of interpretation

Nothing here is a disease model, a drug target validation, or a lifespan intervention. A jellyfish homolog of a human neurodegeneration gene is evidence about the evolutionary depth of that gene's normal function, and a copy-number contrast is a nomination for mechanistic follow-up, not a discovered mechanism. The bridge this project builds between the jellyfish and the human problems runs through shared maintenance biology, stated carefully, with every crossing point labelled by its evidence class. The report's findings stand or fall on the controls; everything beyond them is argument, and the argument is written to be argued with.

### Conclusion and next steps

This expansion set out to do one thing: take the closed phase's calibrated jellyfish genomics and aim it at the two fields the user named, aging and longevity research and neurology. It did that by freezing two new gene panels, adding the chromosome-level mortal-congener genome, reusing the validated mapping pipeline without touching its thresholds, gating everything on positive and negative controls, and pre-registering six hypotheses with their corrections before any outcome data existed.

The deliverables are durable regardless of which way individual hypotheses fell. There is now a gene-by-gene census of the aging and longevity canon in the genome of the only metazoan that repeatedly rejuvenates, and a matching census of the human neurodegeneration gene set, both built on direct genome mapping with recorded checksums, manifests, and ledgers rather than on databases that this organism barely exists in. There is a tested but unresolved copy-number contrast between the immortal species and its mortal relatives. There is a tested statement about whether the aging and neurology panels share their mappable core in this genome. And there is a

negative register that says plainly what did not survive correction, because a screen that cannot fail is not a screen.

The next steps write themselves from the results, and they split by evidence class. For every gene where both *T. dohrnii* assemblies agree and controls are clean, the follow-up is transcriptomic: does the gene's expression move during life cycle reversal, in the published reversal timecourse and in new data? For any candidate copy-number contrast, the follow-up is structural: build the locus properly with long-read data, because copy claims from draft assemblies are provisional by nature. For every absence that matters, the follow-up is sensitivity: longer queries, profile methods, and eventually the annotation effort this organism has always deserved. And for the two fields this report connects, the follow-up is collaborative: the aging community now has a checklist for the limiting-case species, and the neurology community has an evolutionary annotation of its disease-gene list that did not exist before this week.

The immortal jellyfish does not owe anyone a cure. What it owes science, and what this project has collected, is a straight answer to a straight question: what does the genome of the animal that restarts its life actually contain of the machinery that human aging and human neurodegeneration run on? A bounded, search-dependent answer is now on record, gene by gene, with its uncertainties printed next to it. The later whole-proteome screen yielded no validated family expansion; the full PacBio replay could not settle the ATG5 extra-copy question; transcriptomic inference and full raw-data reproduction remain open.

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